6 - 1 MHz oscillator

÷10

÷10

Binary counter

Multiplier

BCD counter

4 - bit comparator

Selector

÷2

÷10

Slate counter

Latch

Select sampling frequency

Sampling circuit

Reconstriction circuits

Second order low pass filter

Ext sampling pulse i/p

Signal input

Sample & Hold amplifier

Sample and hold output

Sixth order low pass filter

Expt. No. 02
Date 02/07/2026

Page No. 31

IMPLEMENT SIGNAL SAMPLING AND RECONSTRICTION

AIM:

To analyse process of sampling and reconstriction using trainer kit.

APPARATUS REQUIRED:

S.No	Apparatus	Range	Quantity
1.	sampling and reconstriction kit	-	1
2.	DSO	(0-30)MHz	1
3.	Power card	-	as required
4.	Patch card	-	as required

PROCEDURE:

Connections are given as per the block diagram

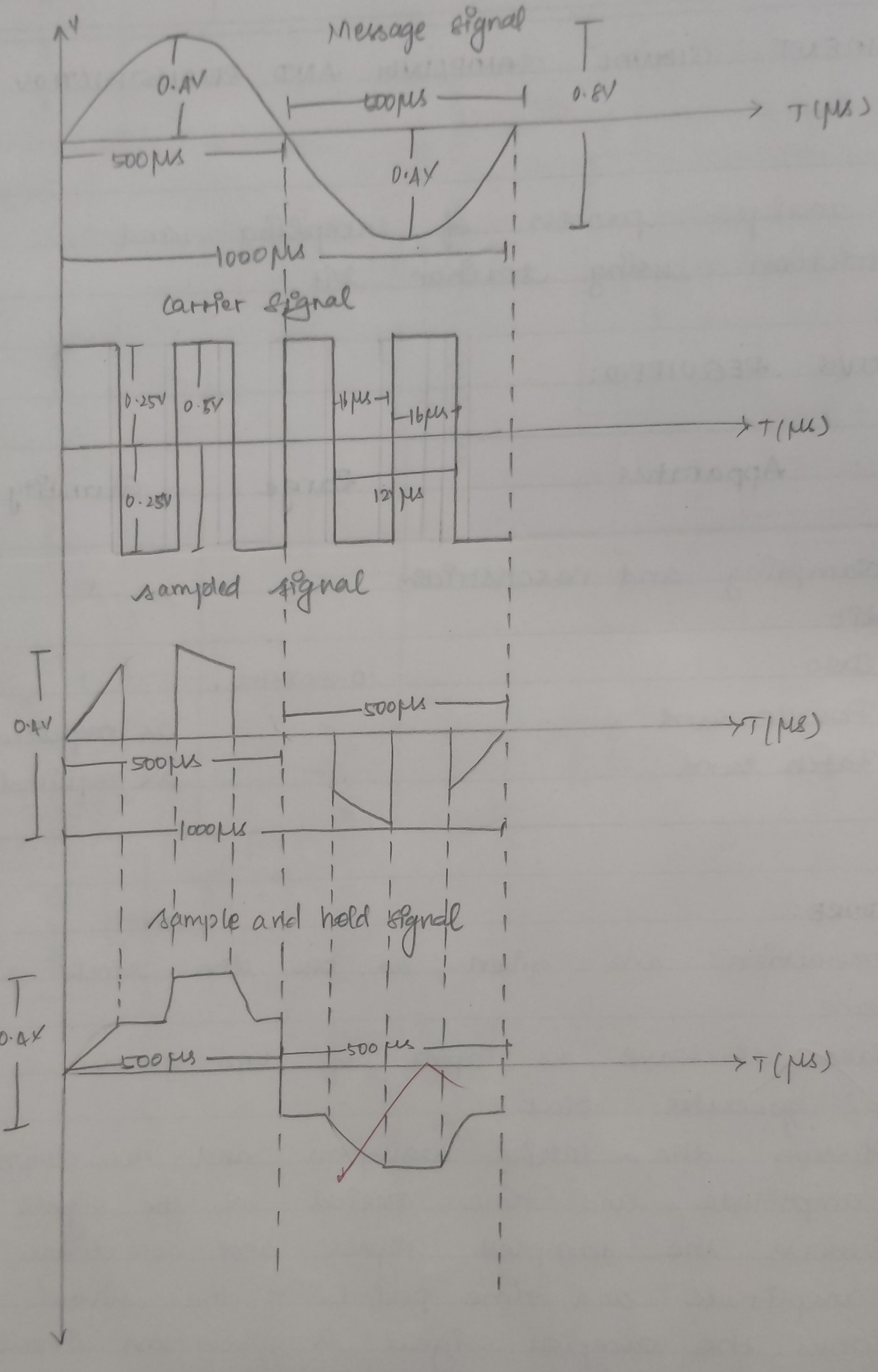
Take sinewave as inputs of 1KHz from signal generator block

Observe the carrier waveform and note down the amplitude and time period of the signal

Observe the sampled signal and note down the amplitude and time period of the signal

Thus the sampled signal is given an input

MODEL GRAPH:


Message signal
V
0.4V
500μs
1000μs
0.8V
T(μs)
0.4V
Carrier signal
0.25V
0.5V
16μs
16μs
T(μs)
0.25V
12μs
sampled signal
0.4V
500μs
T(μs)
sample and hold signal
0.4V
500μs
500μs
T(μs)

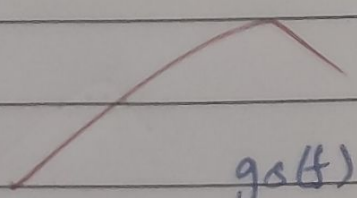
to LPF and the reconstructed waves form obtained in output

THEORY:

Sampling is process of conversion of analog signal to discrete signal sampling shows that a continuous time band limited digital may be presented prefectly by its samples at uniform intervals of T seconds. If this small enough. In other words; the continuous time signal may be reconstructed prefectly from its sample

Sampling Theorem:

It state that the band limited signal of finite energy which has no frequency component higher that 'w'. Fleety is completely describe by specified the values of signal at instant of time separated.



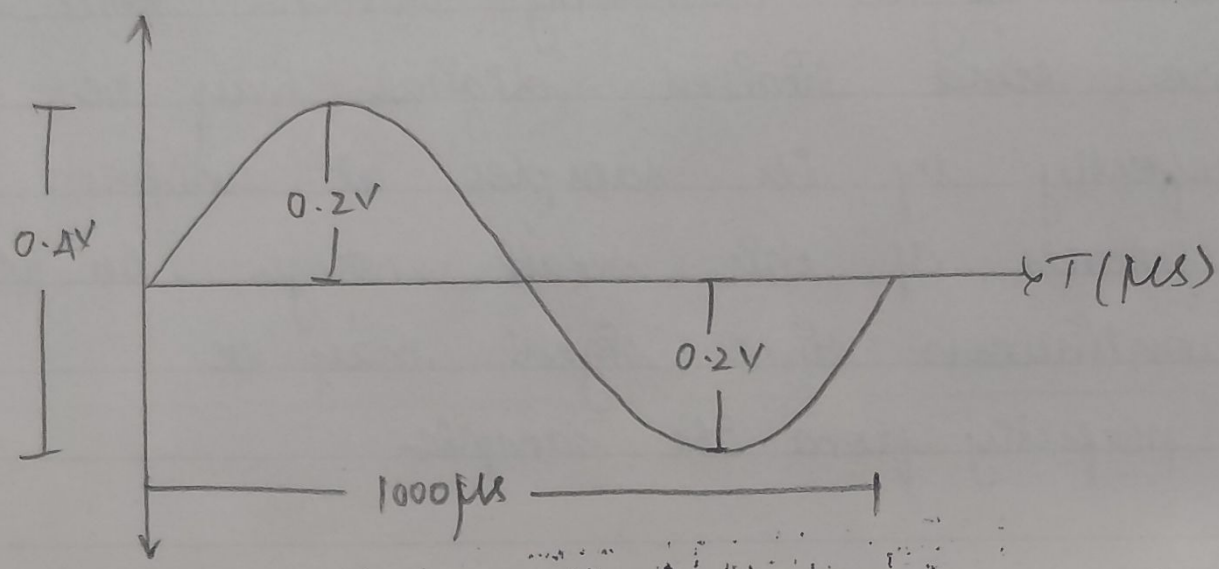
$$g_s(t) = \sum_{n=-\infty}^{\infty} g(nT_s) \cdot \delta(t - nT_s)$$

$g_s(t)$ = The ideal sampled signal

$f_s = \frac{1}{T_s}$; sampling rate

T_s = sampling period.

Reconstructed signal.



Particulars	(Max) Marks	Obtained Marks
Observation	3	2
Output / Results	5	5
Performance /Viva-Voce	2	2
Total	10	09
Sign with date		

RESULT:

Thus the sampling modulation and demodulation is successfully executed.

DIAGRAM:

RF Block
VR2
generator

RF output
TP0
S0

PF Input
TP4
S4

IF Block
VR4
generator

RF output
TP1
S1

Modulation
Block

AM output
TP6
S6

Demod output
S8
TP5

Demodulation
Block

S7
TP7

DSO

Demodulator
O/P

External
I/P
TP3
S3

Ext Block
VR1
5Hz - 50kHz
2KHz AF O/P

TP3
S2

TP5
S5

GND

TG10
SG10

GND

TG12

SG11

GND

Expt. No. 4
Date 09/07/2026

Page No. 39

IMPLEMENT AM - MODULATOR AND DEMODULATOR

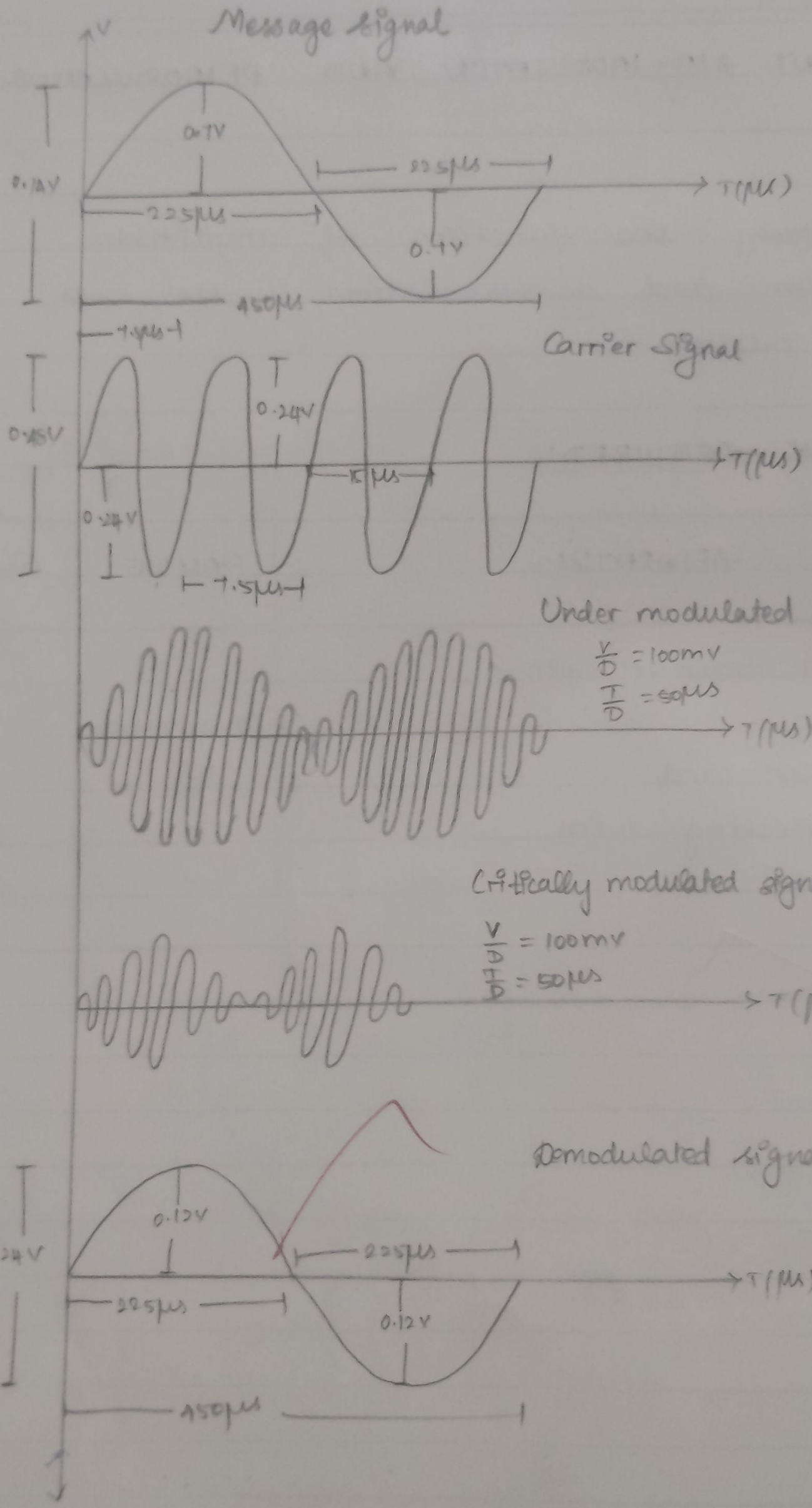
AIM:

To study the function of amplitude modulation and demodulation (under, over, critical modulation)

APPARATUS REQUIRED:

S.NO	APPARATUS	RANGE	QUANTITY
1.	AM trainer kit	-	1
2.	DSO	(0-30) MHz	1
3.	Power card	-	1
4.	connecting wire	-	few
5.	Patch card	-	few

MODEL GRAPH:


Message signal
V
0.1V
0.1V
225μs
225μs
T(μs)
0.1V
450μs
Carrier Signal
0.24V
0.48V
T(μs)
1μs
0.24V
7.5μs
Under modulated signal
V/D = 100mV
T/D = 50μs
T(μs)
Critically modulated signal
V/D = 100mV
T/D = 50μs
T(μs)
Demodulated signal
0.12V
0.24V
225μs
225μs
T(μs)
0.12V
450μs

Expt. No.
Date

Page No. 89

PROCEDURE:

Switch on the trainer kit and check the output of the carrier generator on oscilloscope

Connect ground 1KHz with 2 volts, AF signals at AF input to the modulator circuit

Connect the carrier signal at the input of the modulator circuit

Observe the modulator output signal at an output

Vary the modulating frequency and amplitude and observe the effect on the modulated waveform

The depth of modulation can be varied using variable knob provided at AF input

The percentage modulation can be calculated using formula.

THEORY:

MODULATION:

Modulation is defined as process changing the characteristics of the carrier signal in accordance with intensity of message signal

Amplitude modulation is defined as a system of modulation in which the amplitude of carrier is varied. In accordance with

signal remain constant.

$m(t) = A_m \cos \omega_m t$

$c(t) = A_C \cos \omega_c t$

$x_c(t) = (A_C + A_m \cos \omega_m t) \cos \omega_c t$.

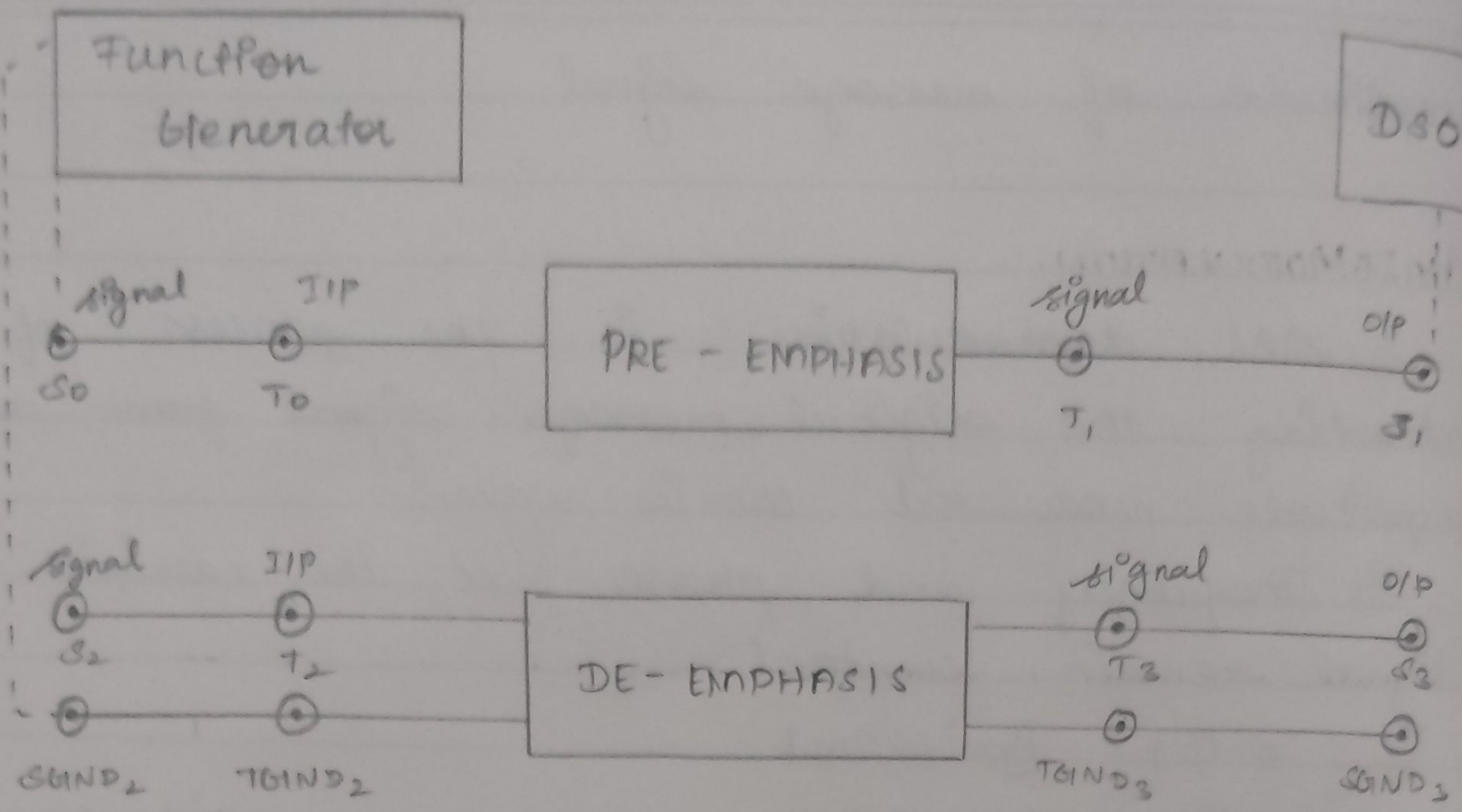
Particulars	Max Marks	Obtained
Observation	3	3
Output / Results	5	5
Performance /Viva-voce	2	2
Total	10	10
Signature with date		

RESULT:

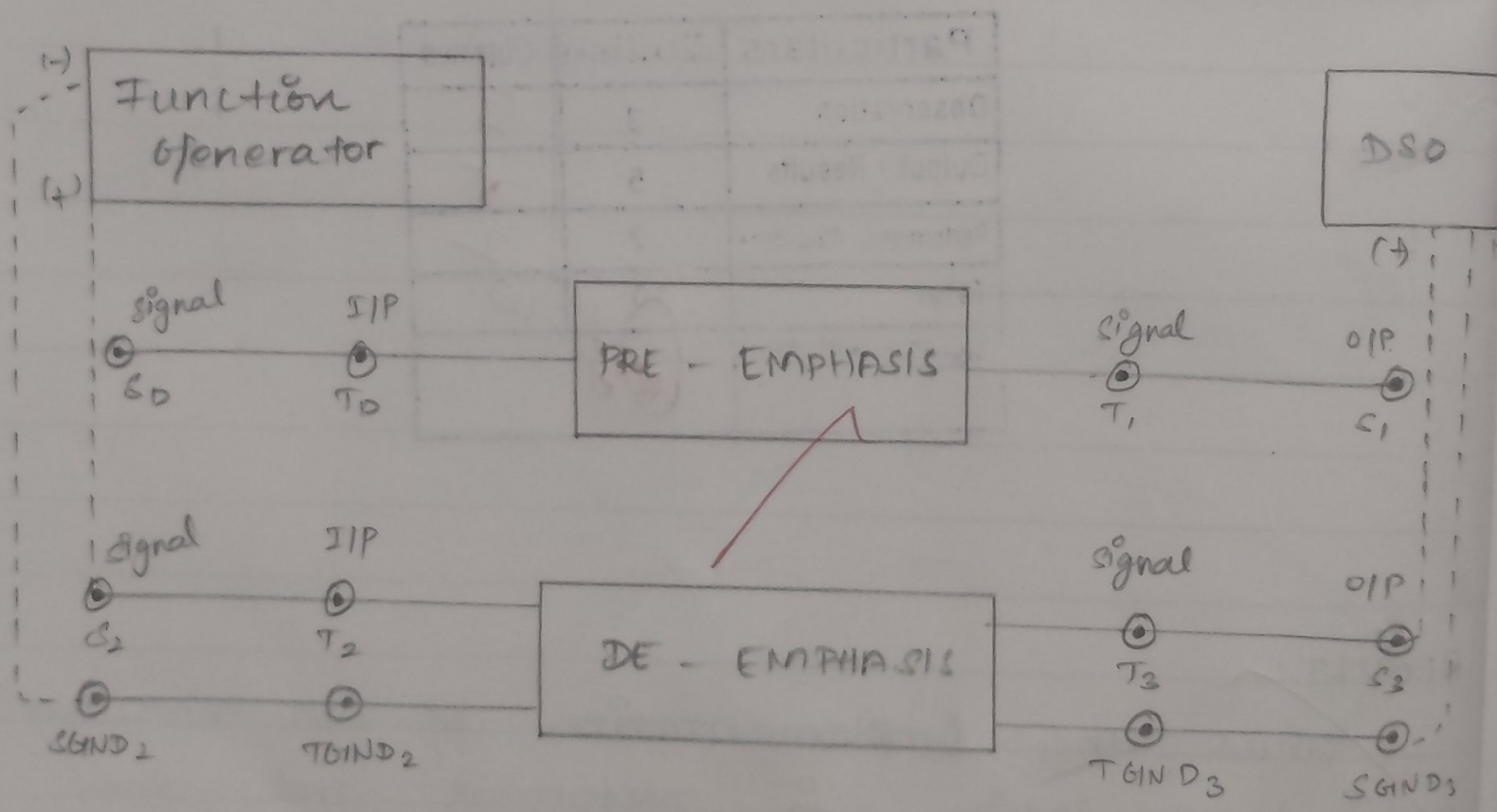
Thus the implementation of AM modulation and demodulation is performed and the output is verified

DIAGRAM:

PRE - EMPHASIS:



DE - EMPHASIS:



Expt. No. 6
Date 16/07/2026

Page No. 13

IMPLEMENT PRE-EMPHASIS AND DE-EMPHASIS.

AIM:

To perform pre-emphasis and de-emphasis using the trainer kit

APPARATUS REQUIRED:

S.NO	APPARATUS	RANGE	QUANTITY
1.	Pre-emphasis and De-emphasis kit	-	1
2.	Function Generator	(0-30) MHz	1
3.	DSO	(0-30) MHz	1
4.	Power Cord	-	1
5.	Patch cord	-	few

TABULATION:

PRE-EMPHASIS:

$V_{LR} = 2V$

S.No	Frequency (Hz)	Volts (V_o)	$20 \log \left(\frac{V_o}{V_i}\right)$
1.	100	2.04	0.17
2.	200	2.10	0.42
3.	300	2.30	1.213
4.	400	2.56	2.14
5.	500	2.82	2.98
6.	600	3.14	3.91
7.	700	3.46	4.76
8.	800	4.76	5.51
9.	900	5.57	6.23
10.	1K	6.23	6.84
11.	2K	6.84	10.60
12.	3K	10.60	12.94
13.	4K	12.94	13.57
14.	5K	13.57	14.15
15.	6K	14.15	14.15
16.	7K	14.15	14.15
17.	8K	14.15	14.15
18.	9K	14.15	14.15
19.	10K	14.15	14.15

Expt. No.
Date

Page No. 45

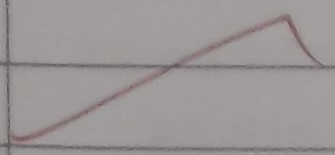
PROCEDURE:

↳ Pre-emphasis:

i) Switch on the trainer kit
ii) Give the input signal from function generator to pre-emphasis
iii) Observe the output using DSO
iv) By varying the input frequency note down the output.

↳ De-emphasis:

i) give the output of the pre-emphasis to input of the de-emphasis
ii) Observe the output signal
iii) The resonance will similar to low pass filter

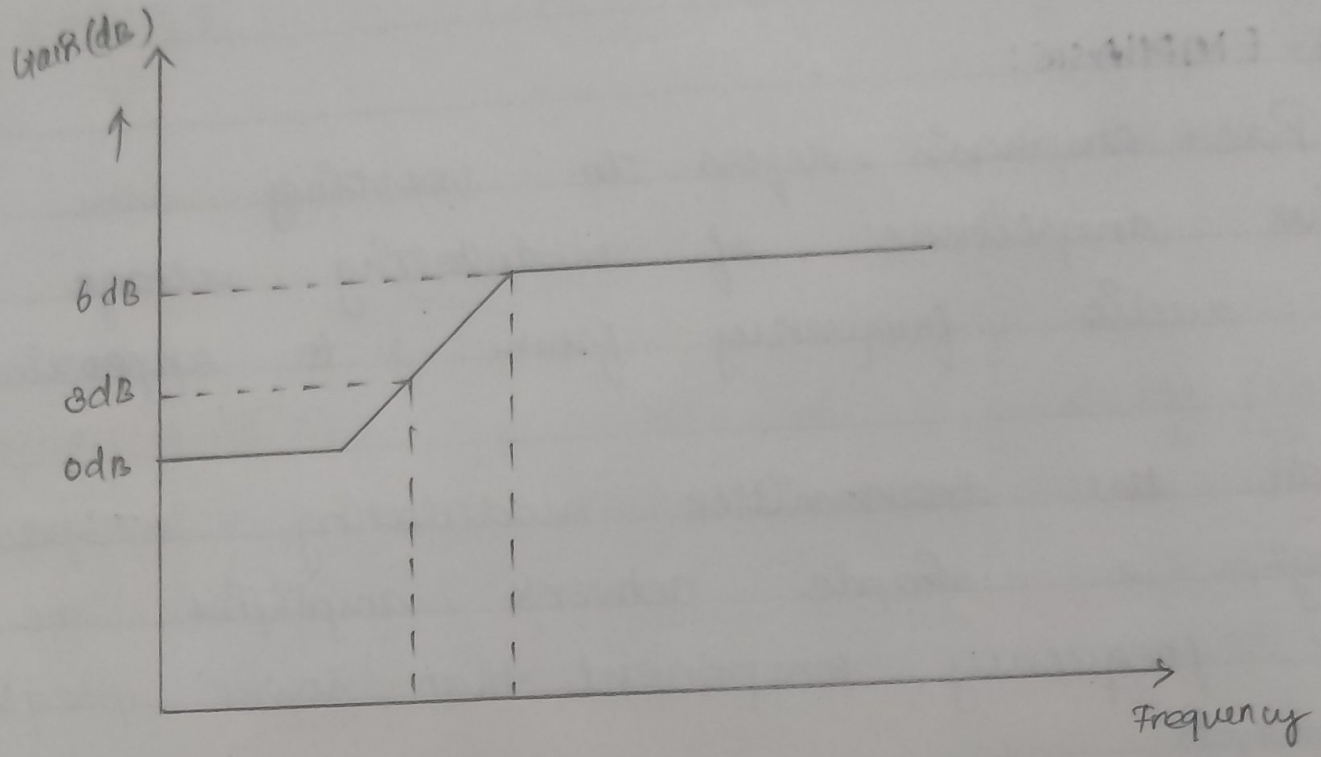


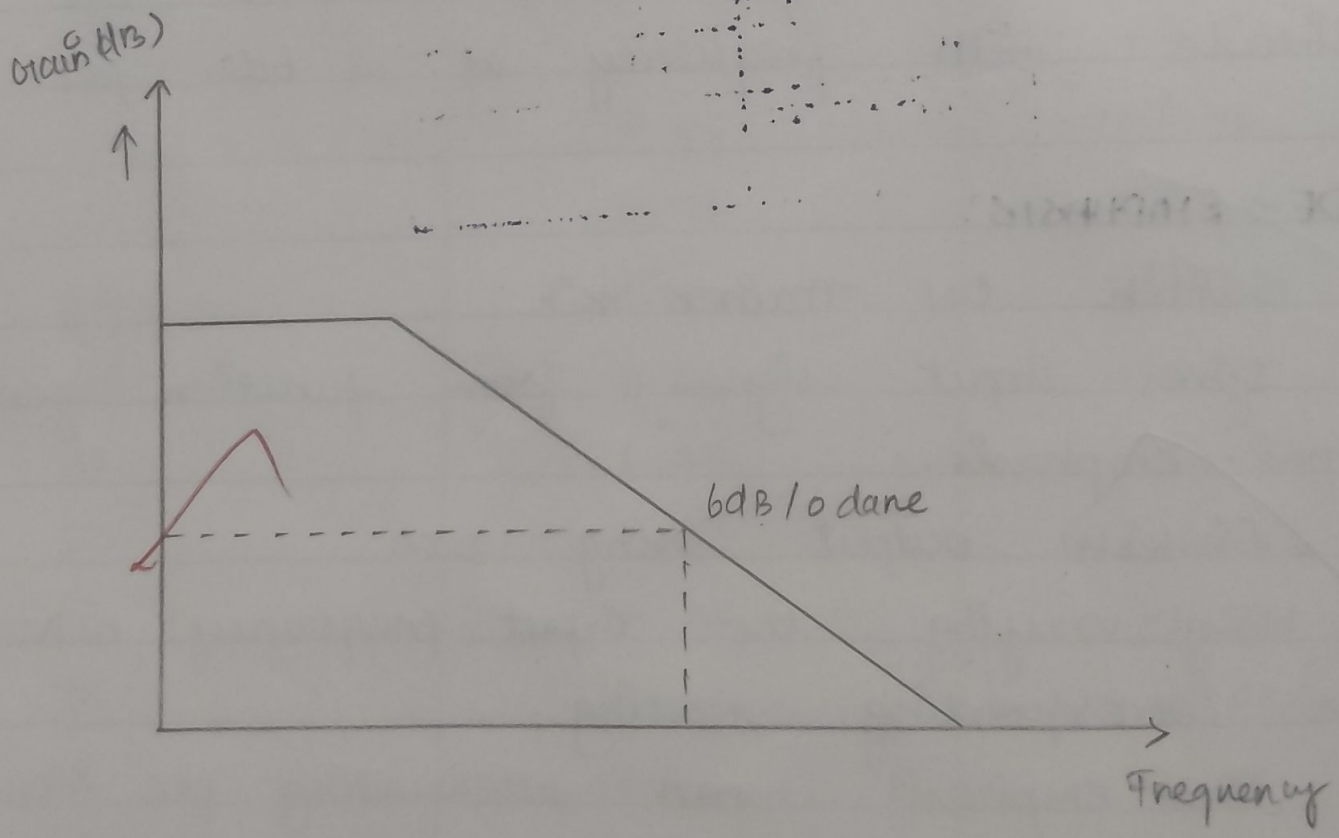
DE-EMPHASIS :

$V_{in} = 2V$

S.No	Frequency (Hz)	Volts (V_o)	$20 \log \left[\frac{V_o}{V_i}\right]$
1.	100	1.92	-0.35
2.	200	1.68	-1.51
3.	300	1.48	-2.61
4.	400	1.32	-3.60
5.	500	1.20	-4.43
6.	600	1.06	-5.51
7.	700	0.96	-6.37
8.	800	0.90	-6.93
9.	900	0.84	-7.57
10.	1K	0.76	-8.40
11.	2K	0.50	-12.44
12.	3K	0.44	-13.15
13.	4K	0.40	-13.97
14.	5K	0.38	-14.43
15.	6K	0.36	-14.89
16.	7K	0.36	-14.89
17.	8K	0.36	-14.89
18.	9K	0.36	-14.89
19.	10K	0.36	-14.89

MODEL GRAPH:





Particulars	Max. Marks	Obtained
Observation	3	3
Output / Results	5	5
Performance /Viva-Voce	2	2
Total	10	10
Signature with date		

RESULT:

Thus the output waveform for pre-emphasis and de-emphasis are generated and output also verified successfully.

Diagram:

clock generator divider

64kHz

32kHz

16kHz

8kHz

CLK rate selector switch

500Hz

1kHz

2kHz

Square Cosine

Non - sinusoidal source

IF amplifier

Variable Gain

DELTA MODULATOR

Demodulation

variable gain

VR4

4th order LPF

3rd order LPF

TP1

TP2

TP3

TP4

TP5

TP6

TP7

TP8

TP9

TP10

TP11

TP12

TP13

TP14

TP15

TP16

TP17

TP18

TP19

TP20

TP21

TP22

TP23

TP24

TP25

TP26

TP27

TP28

TP29

TP30

TP31

TP32

TP33

TP34

TP35

TP36

TP37

TP38

TP39

TP40

Expt. No. 6.
Date 13/05/2026

Page No. 51

IMPLEMENTATION OF DELTA MODULATION AND DE-MODULATION

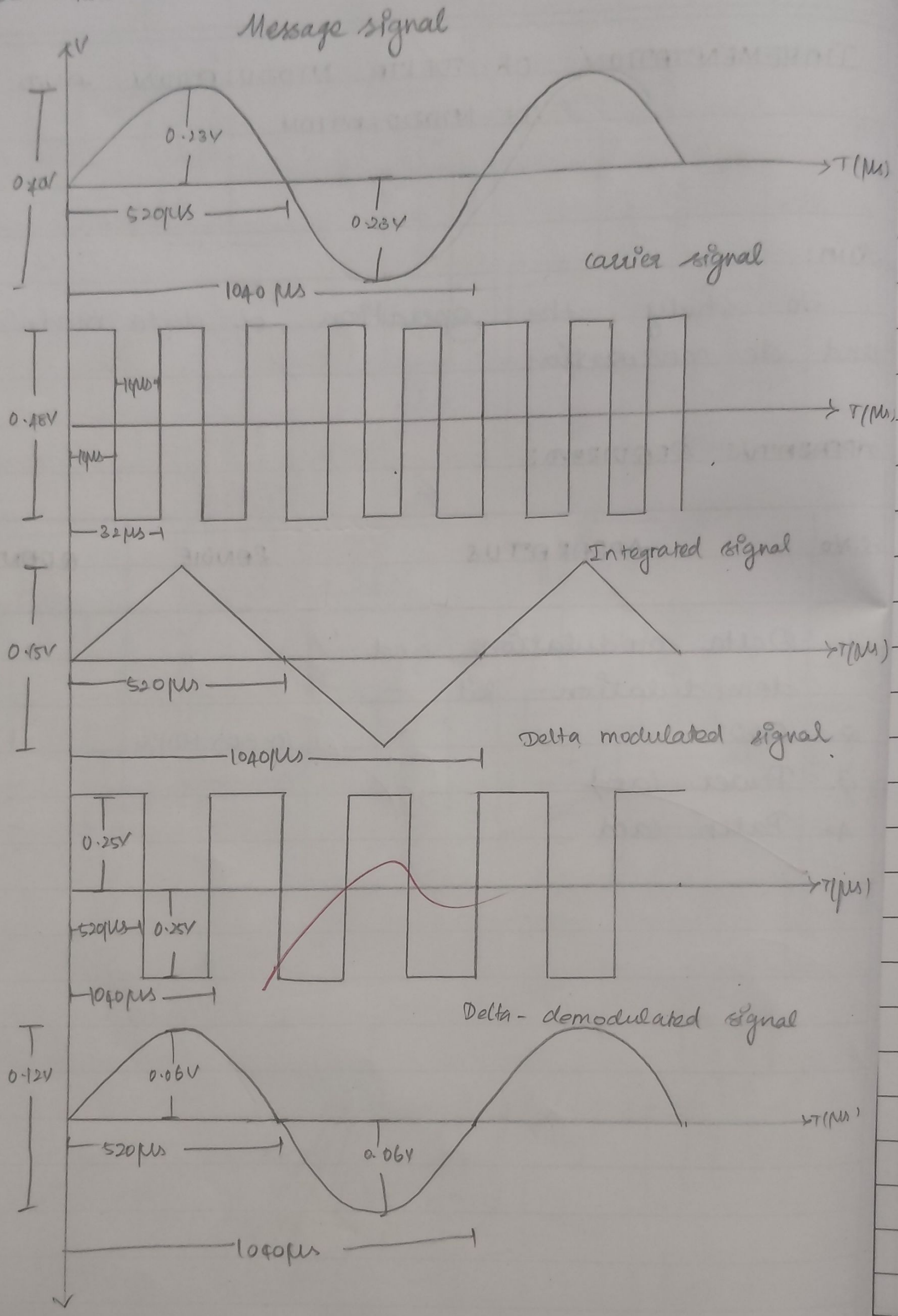
AIM:

To study the operation of delta modulation and de-modulation

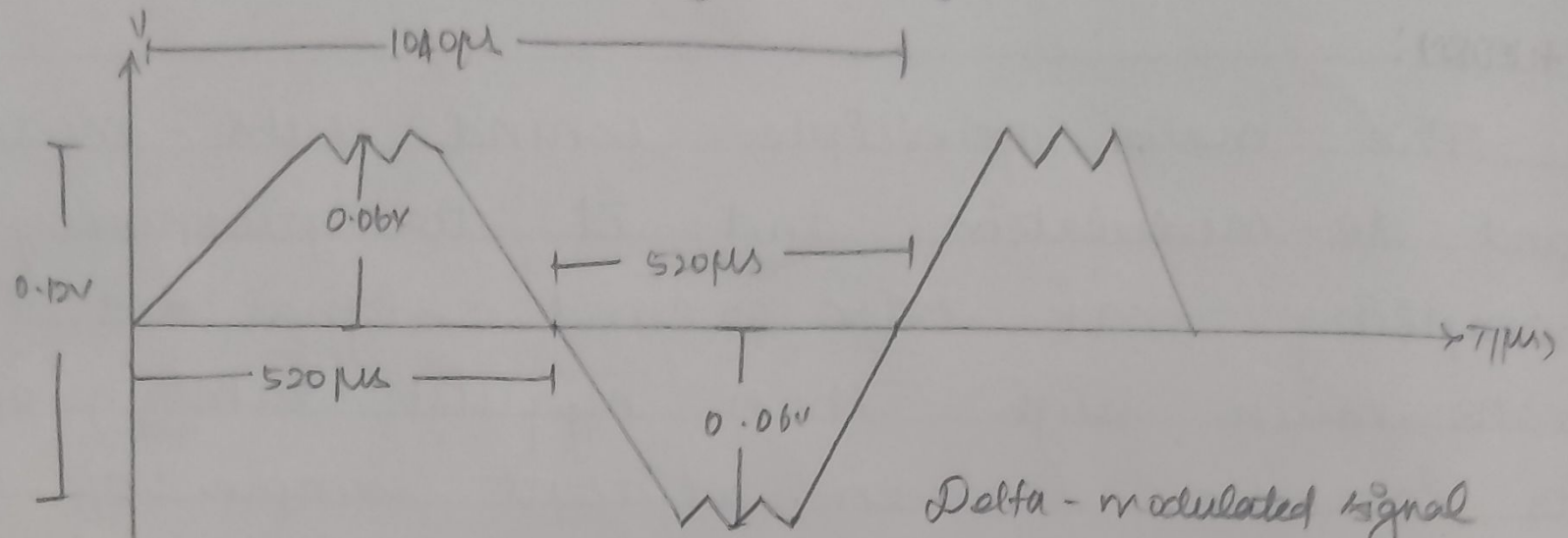
APPARATUS REQUIRED:

S.No	APPARATUS	RANGE	QUANTITY
1.	Delta modulation and demodulation kit	–	1
2.	DSO	(0-30) MHz	1
3.	Power card	–	–
4.	Patch card	–	–

MODEL GIRAPH:

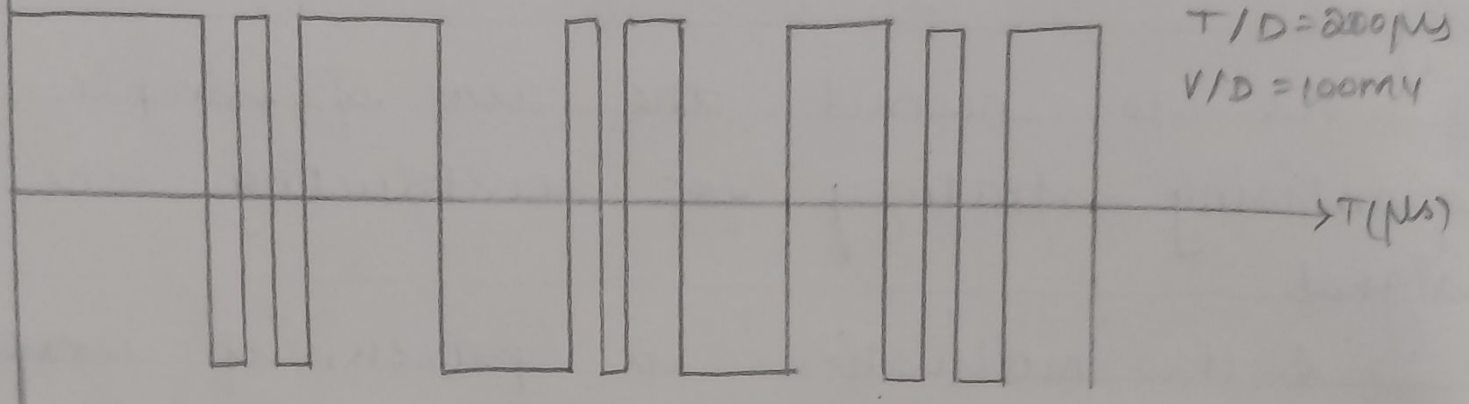


Integrated signal

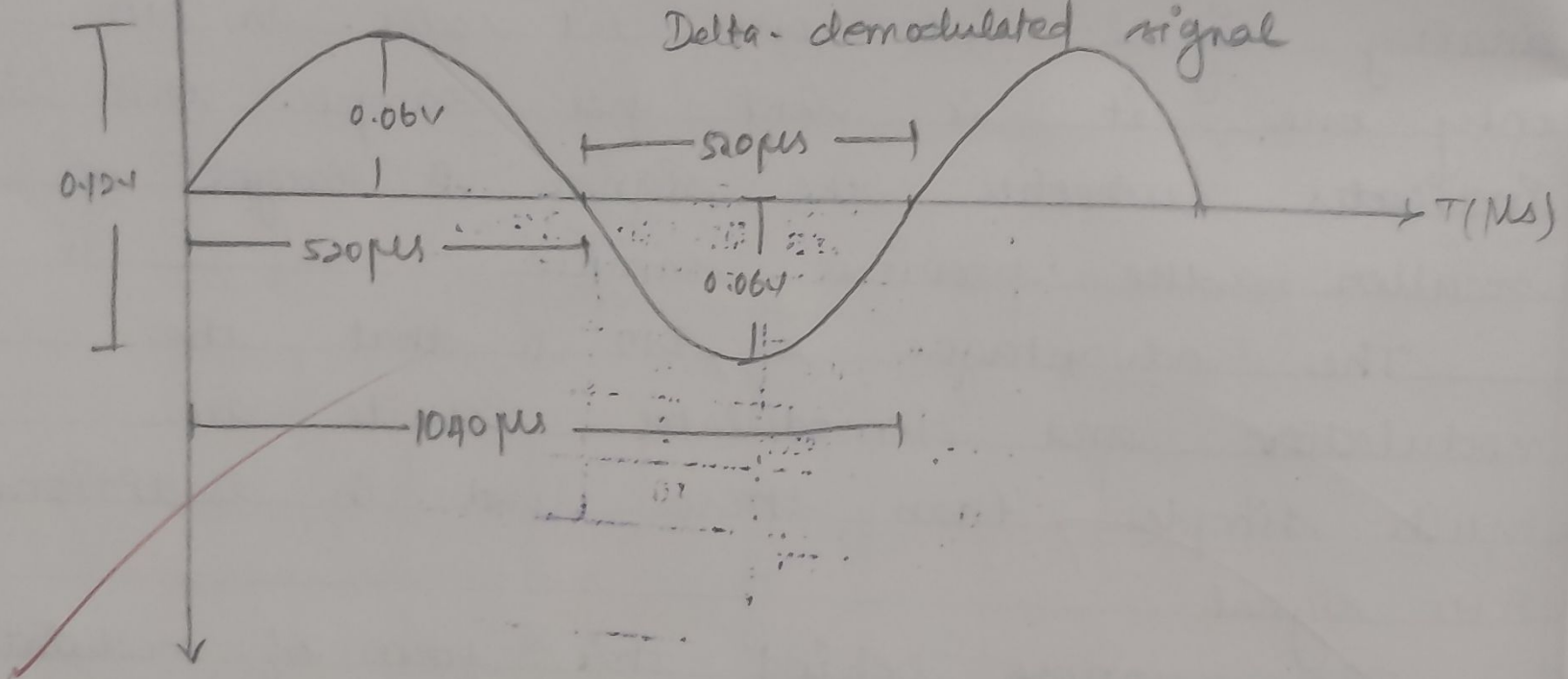


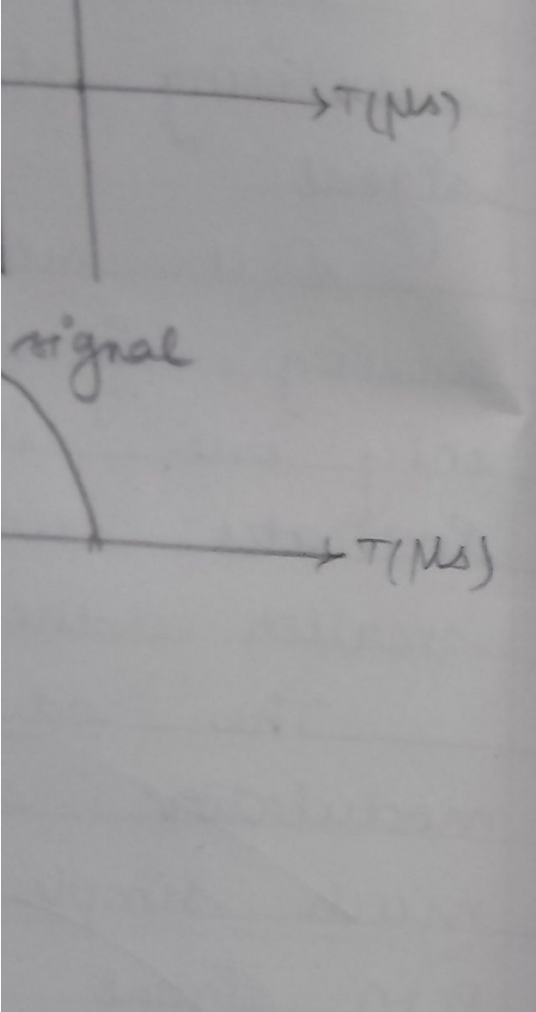
Delta - modulated signal

T/D = 200μs
V/D = 100mV



Delta - demodulated signal





trace oscilloscope on channel

Connect clock signal to the DM circuit

Observe the DM output to the Delta modulation

Connect DSO output to the channel

Observe the demodulating output with and without PC filter or CRO.

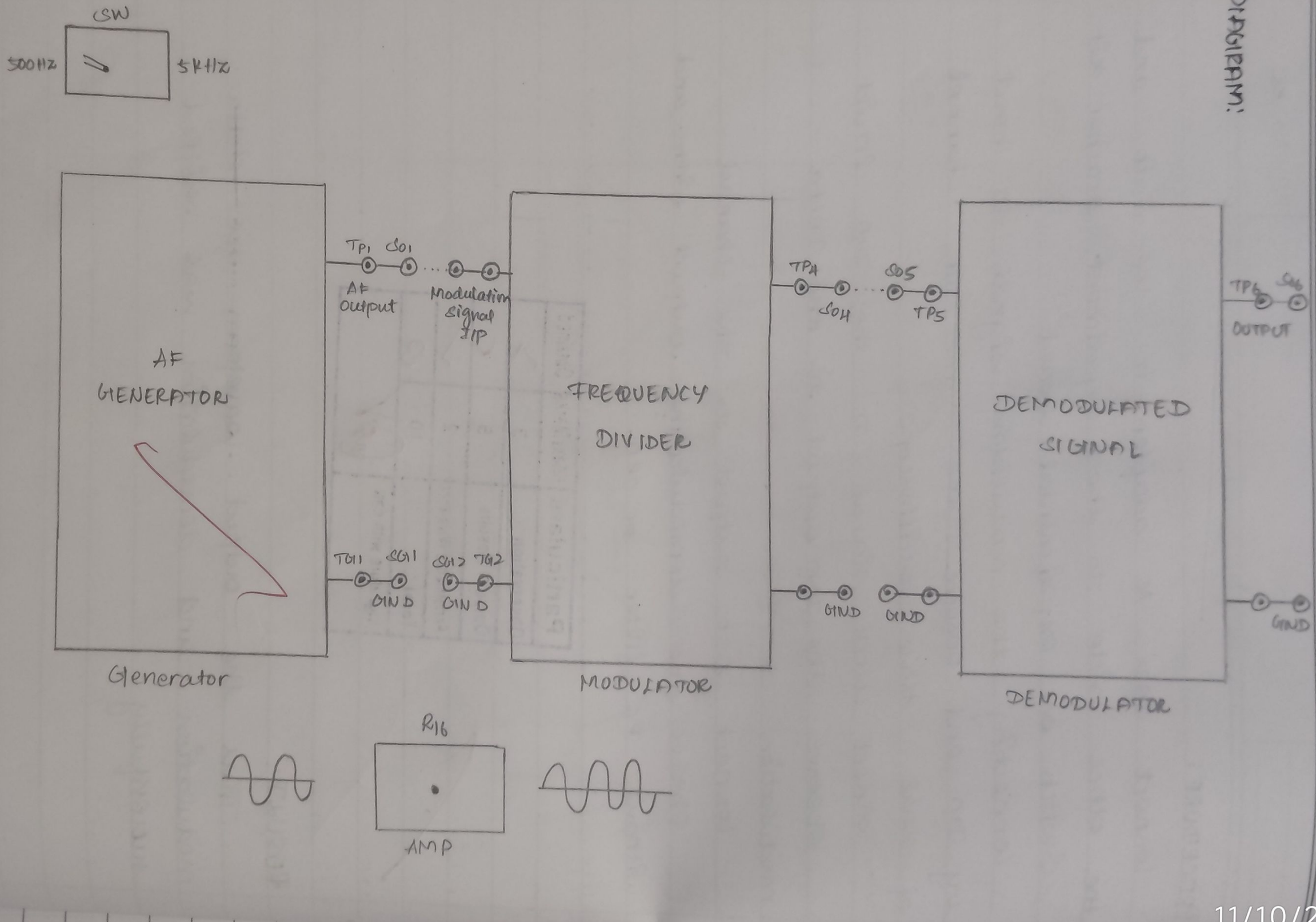
Particulars	(Max) Marks	Obtained
Observation	3	3
Output / Results	5	5
Performance /Viva-Voce	2	✓
Total	10	10
Signature with date		

RESULT:

Thus the output waveform for delta modulation and demodulation was waitted sucessfully.

OCO M6 PRO 5G

DIAGRAM:


SW
500Hz
5kHz
AF GENERATOR
Generator
TP1
SO1
AF output
Modulation signal I/P
FREQUENCY DIVIDER
MODULATOR
TP4
SO4
SO5
TP5
DEMODULATED SIGNAL
DEMODULATOR
TP6
SO6
OUTPUT
TG1
SG1
GND
SG2
TG2
GND
GND
GND
GND
R16
AMP

Expt. No. 7.
Date 21/08/2026

Page No. 57

IMPLEMENTATION OF FM MODULATOR AND DEMODULATOR

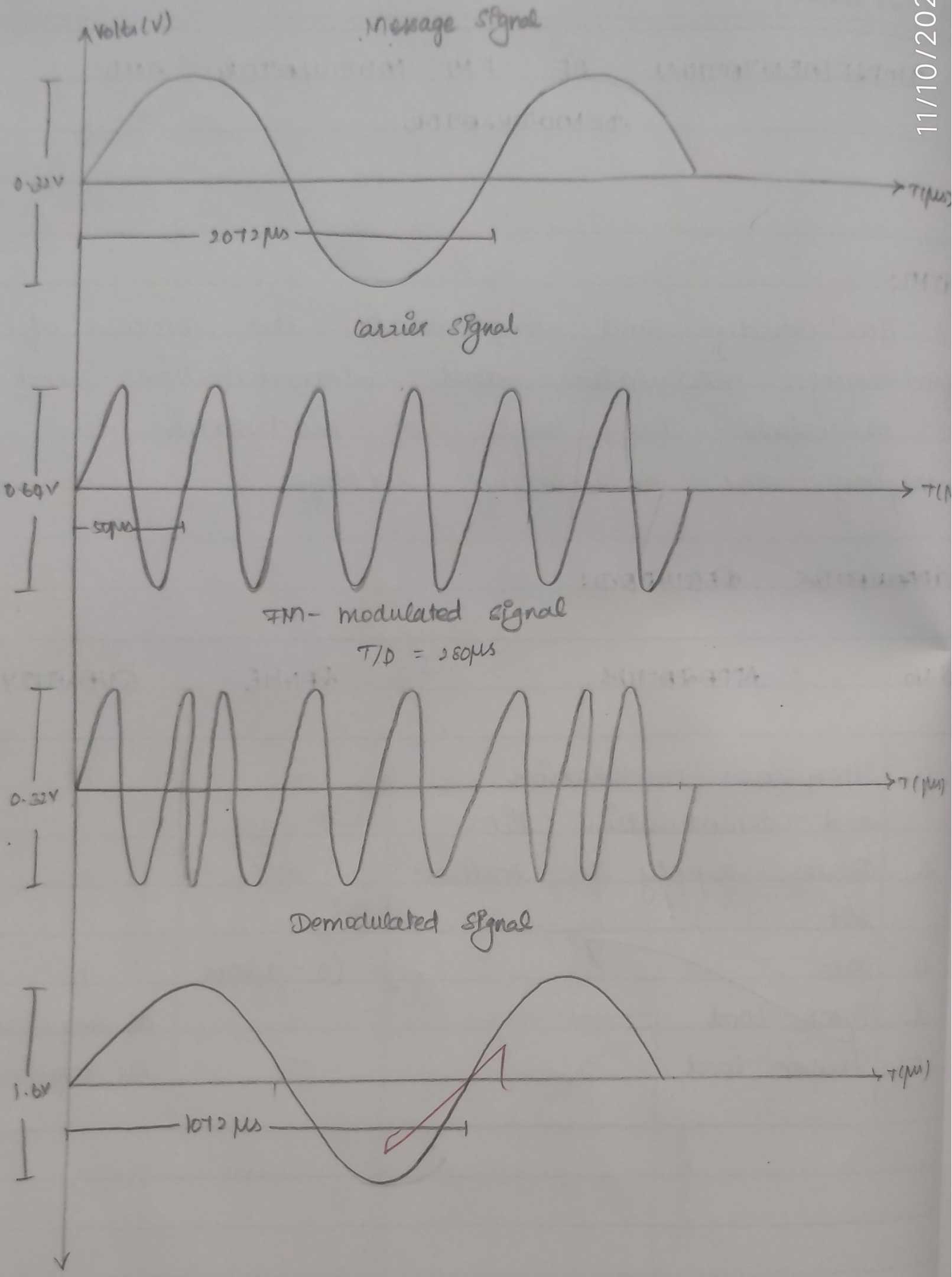
AIM:

To study and implement the process of frequency modulation and demodulation and to calculate the length of modulation by varying the modulating voltage.

APPARATUS REQUIRED:

S.NO	APPARATUS	RANGE	QUANTITY
1.	Frequency modulation and demodulation kit	-	1
2.	Power supply for trainer kit	-	1
3.	DSO	(0 - 30)MHz	1
4.	Patch cord	-	As required
5.	Power cord	-	As required

MODEL GRAPH:


Volts(V)
Message Signal
0-32V
2072μs
T(μs)
Carrier Signal
0.60V
50μs
T(μs)
FM- modulated signal
T/D = 280μs
0-32V
T(μs)
Demodulated signal
1.6V
1012 μs
T(μs)

into corresponding voltage variations

In many FM systems, PLL-based demodulation is used, a PLL can track the instantaneous frequency of FM signal and produce a corresponding output voltage. The recovered signal should have approximately the same frequency and waveform as the original AF signal, although its amplitude may differ.

PROCEDURE:

FM MODULATOR:

1. Switch OFF the trainer kit
2. Connect the trainer kit to the specified power supply and switch it on.
3. Connect the AF output of AF generator (i.e) TP1 to the modulating signal input
4. Connect the corresponding GND terminals
5. Connect the AF output to CRO channels and

1. Connect the modulated signal output TP4 / SO4 to the input TP5 / SO5 of FM demodulator.
2. Connect the corresponding GND terminals
3. Connect the output of FM demodulator to DSO
4. Observe the demodulated waveform
5. Adjust the frequency - deviation control if required to obtain a clear output
6. Measure and record the frequency of the recovered signal
7. Verify that the frequency of demodulated signal is approximately equal to frequency of original modulating signal

RESULT:

Thus the frequency modulation and demodulation was implemented and the original AF signal was recovered successfully by demodulation.

BLOCK DIAGRAM:

Expt. No. 8
Date 03/09/26

Page No. 65

IMPLEMENT PAM, PPM AND PWM MODULATION AND DEMODULATION:

AIM:

To study and implement pulse amplitude modulation [PAM], Pulse width modulation [PWM], and Pulse Position modulation [PWM] and their corresponding demodulation techniques

APPARATUS REQUIRED:

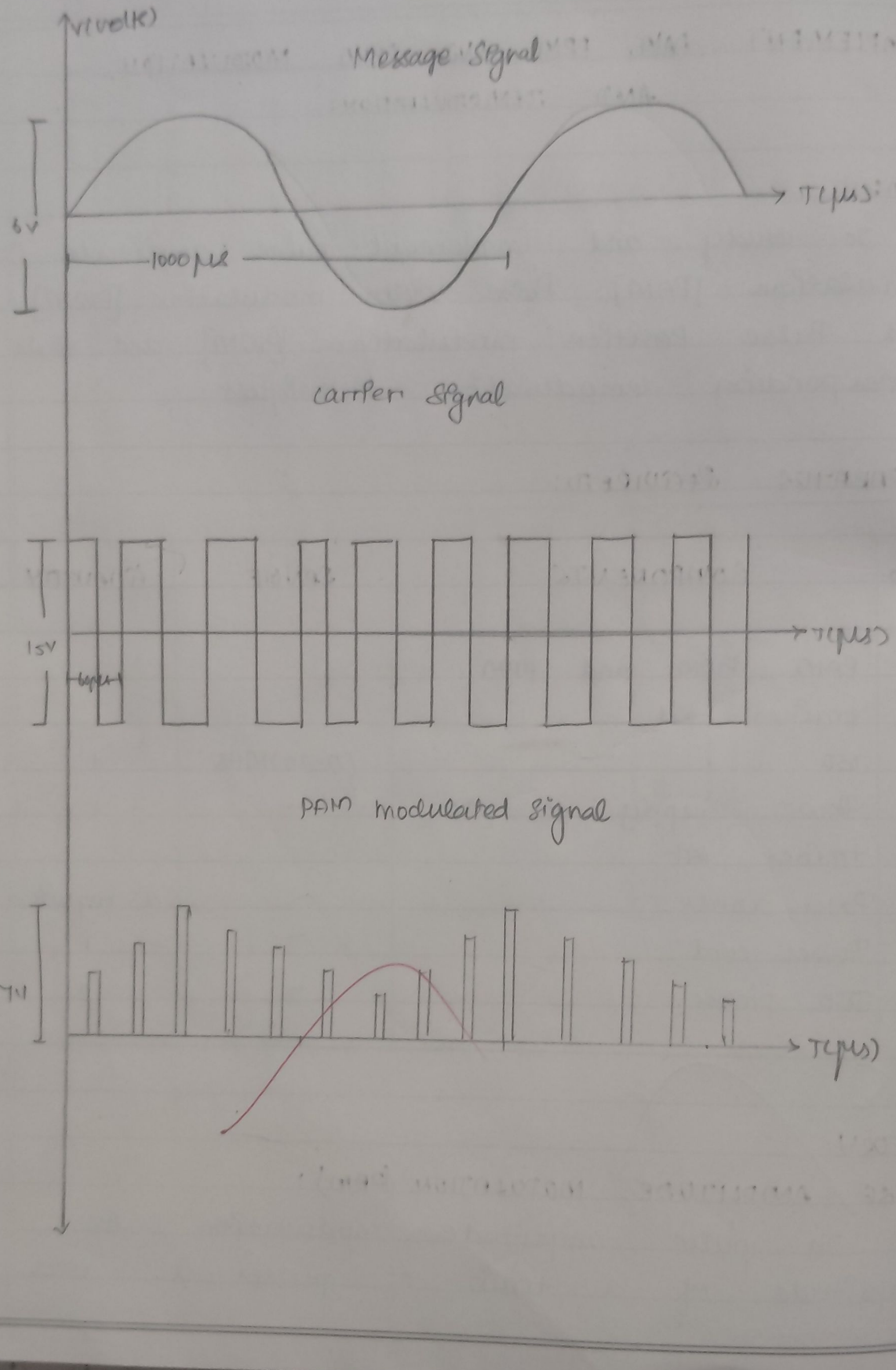
S.No	COMPONENTS	RANGE	QUANTITY
1.	PAM, PWM and PPM trainer kit	–	1
2.	DSO	(0-30)MHz	1
3.	Power supply for the trainer kit	–	1
4.	Patch cords	–	as required
5.	Power cord	–	1
6.	DSO probes.	–	2

THEORY

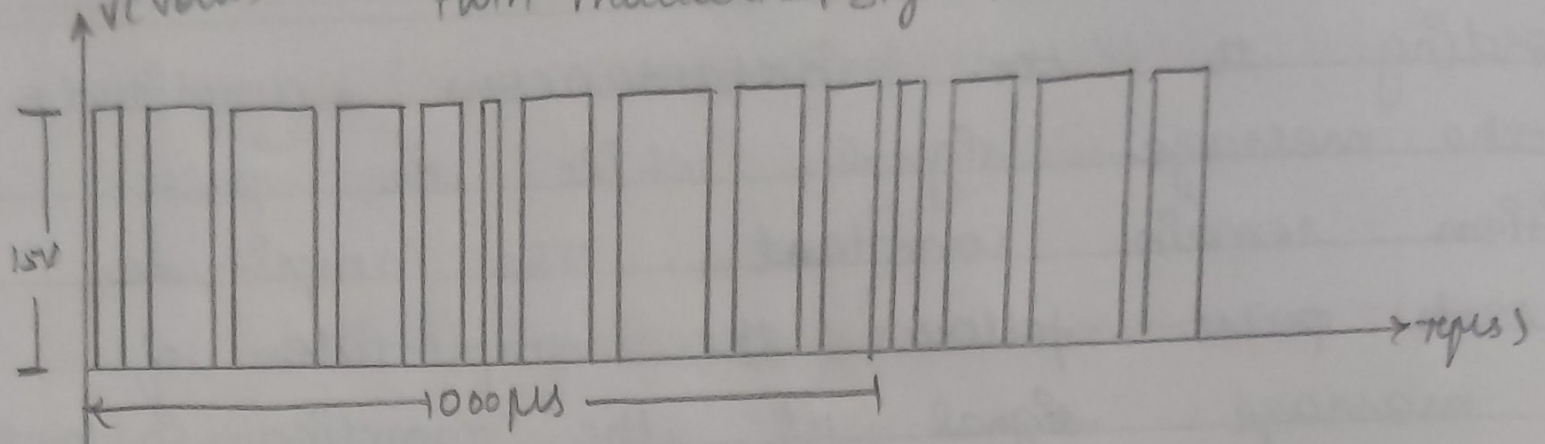
PULSE AMPLITUDE MODULATION (PAM):

In pulse amplitude modulation, the amplitude of a train of pulses is varied

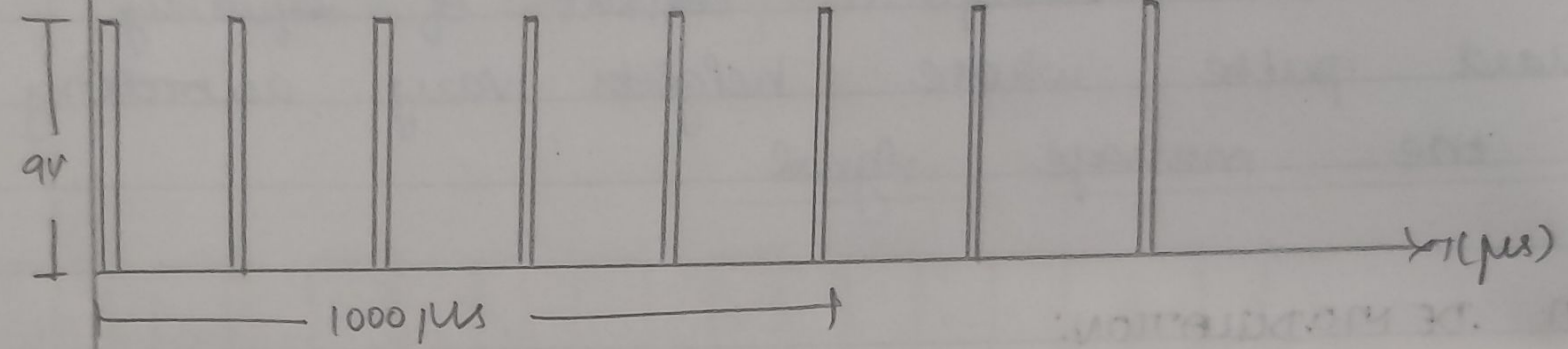
MODEL GRAPH:


V(volt)
Message Signal
6V
1000μs
T(μs)
carrier Signal
15V
T(μs)
PAM modulated signal
7V
T(μs)

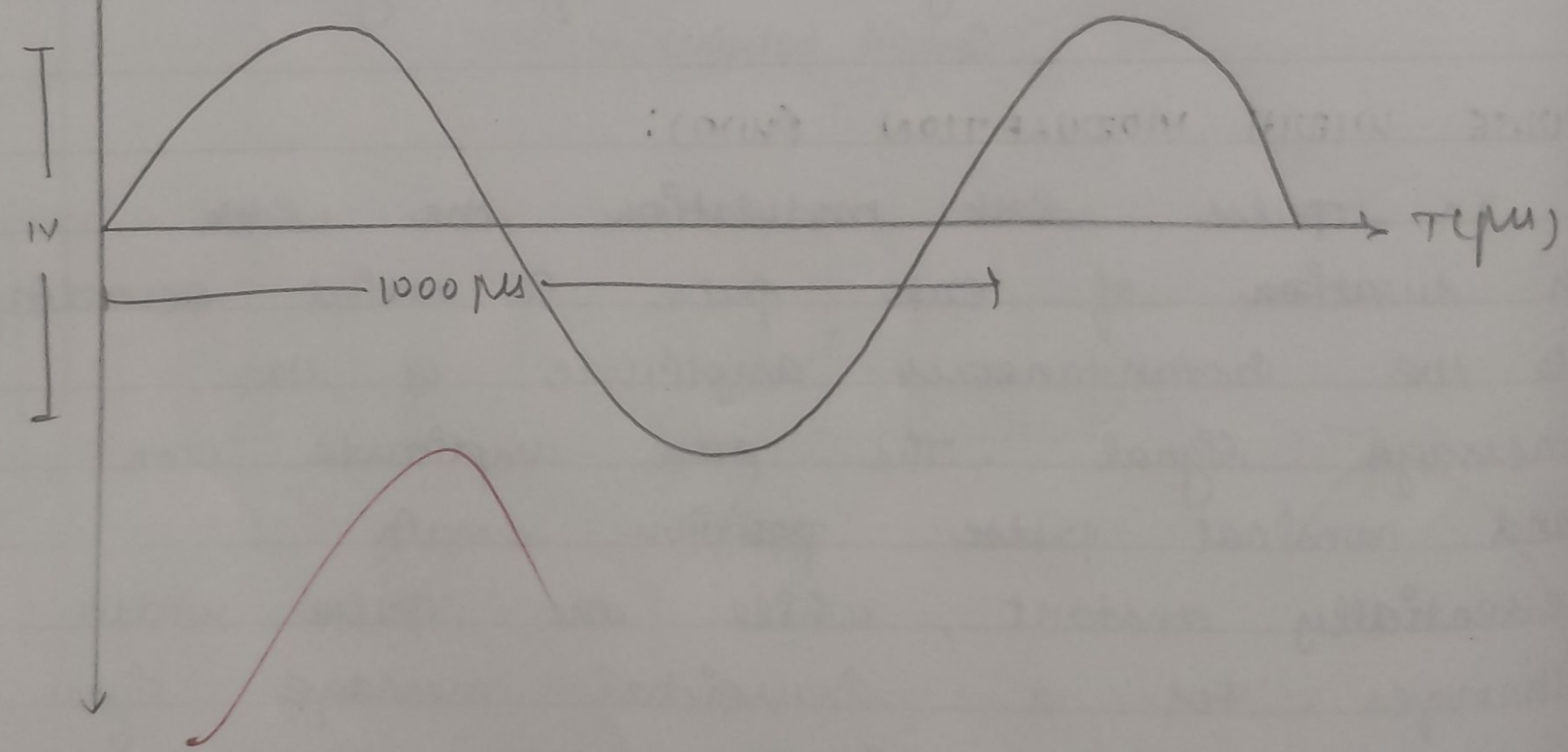
PWM modulated signal



PPM modulated signal



Demodulated signal



PPM DEMODULATION:

The PPM signal is first converted into a suitable PWM / PDM - type signal using a synchronized circuit, and then passed through a low-pass filter to recover the original message signal

PROCEDURE:

1) Switch OFF the trainer kit and make the required connections using the patch cords according to the trainer - kit block diagram
2) Connect the message signal from the signal source section to the input of the required pulse - modulation section
3) Connect the required sampling / clock / carrier signal to the modulator.
4) Switch on the trainer kit.

5) Observe the message signal and sampling / clock signal on the DSO and adjust the amplitude and frequency to obtain stable waveforms

6) Observe the corresponding modulator output at the specified test point

7) Verify the modulation
   ↳ PAM - Pulse amplitude varies
   ↳ PWM - Pulse width varies
   ↳ PPM - Pulse position varies

8) Connect the modulated output to the corresponding demodulator / reconstruction section of trainer kit

9) Observe the demodulator output on the DSO

10) Adjust the demodulator LPF control, if provided, to obtain a clear and stable waveform.

11) Compare the demodulated output with the original message signal.

RESULT:

The PAM, PWM and PPM signals were successfully generated and demodulated, and the original message signal was approximately recovered.

PCM Transmitter

Intelligent signal

500Hz

S0 — TP1

S1 — TP2

GND1 — GND2

DC source

DC-1

S2 — TP3

DC-2

S3 — TP0

GND3 — GND4

Mode of operation

Fast mode

Slow mode

PRBS Generator

D0 D1 D2 D3

TP42 TP44

TP31 LD1

TP30 L5 Tx SINE

PRBS O/P

S5 TP5 ÷1

S4

S6 TP6 ÷1

S7

TP11 ÷1

Audio in

TPLA1 S8 AUDIO TPLA S9

MIC

GND6 GND7

Shift Register

TP16 TP17 TP18 TP19 TP20 TP21

TPA6 TPA7

DP0 DP1

Error Check Code Generator

TP28 TP29 TP25 TP26 TP27 TP24 TP31

TP32 TP33

DP3

ND Converter

TP30 TP31 TP32 TP33 TP34 TP35 TP36 TP37

MSB

Output logic

TP45 Remote

D4 D44

GND

GND5

NONE

Hammy

Switched Faults

11/10/2026

Expt. No. 1e
Date 24/9/26

Page No. 105

IMPLEMENT PULSE CODE MODULATION AND DEMODULATION

AIM:

To convert an analog signal into a pulse digital signal using PCM modulated system and to convert the digital signal into analog signal using PCM demodulation system.

APPARATUS REQUIRED:

S.No	APPARATUS	RANGE	QUANTITY
1.	PCM modulation and demodulation trainer kit	-	1
2.	DSO	(0-30) MHz	1
3.	Power cord	-	1
4.	Patch cord	-	1
5.	Power supply	-	1

THEORY

PCM MODULATION:

Pulse code modulation (PCM) is a form of digital to analog conversion in which the information contained in the samples of an analog signal can be represented

Rx clk

TP3

L8

TP

L9

Rx data

TP2

Reset

SW2

Rx data

TP1

GND

PCM O/P

TP 32

Rx data

Phase detect

Clock O/P

TP31

Data in

TP 6A

TP AA

Data out

PRBS Generator / Detector

L1 L2 L3 L4 L5

TP33 TP34 TP35 TP36

Sync clk Indicator TP44

Sync O/P

Sync O/P TP42

Audio amplifier

SO

TP

CN5

VR1

L10 TP37 TP38 L11

TP39 LPF-1

TP45

CH0

S1

TP40 LPF-2 CH1

TP46

L12

L20 L21 L22 L23 L24 L25 L26

TP30 TP29 TP28 TP27 TP32 TP33 TP34

RP7

RP6

L12 L14 L15 L16 L17 L18 L19

TP8 TP5 TP6 TP7 TP8 TP9 TP10

RP1

RP2

Shift

Register

L26 L24 L28 L29 L30 L31 L32

TP3 TP11 TP12 TP13 TP14 TP15 TP16

Error detection correction

L34 L35 L37 L36 L38 L40

Hamming bits

Error parity

L41

Error code Select

None

odd parity

Error parity

Hamming

SW1

ON SP

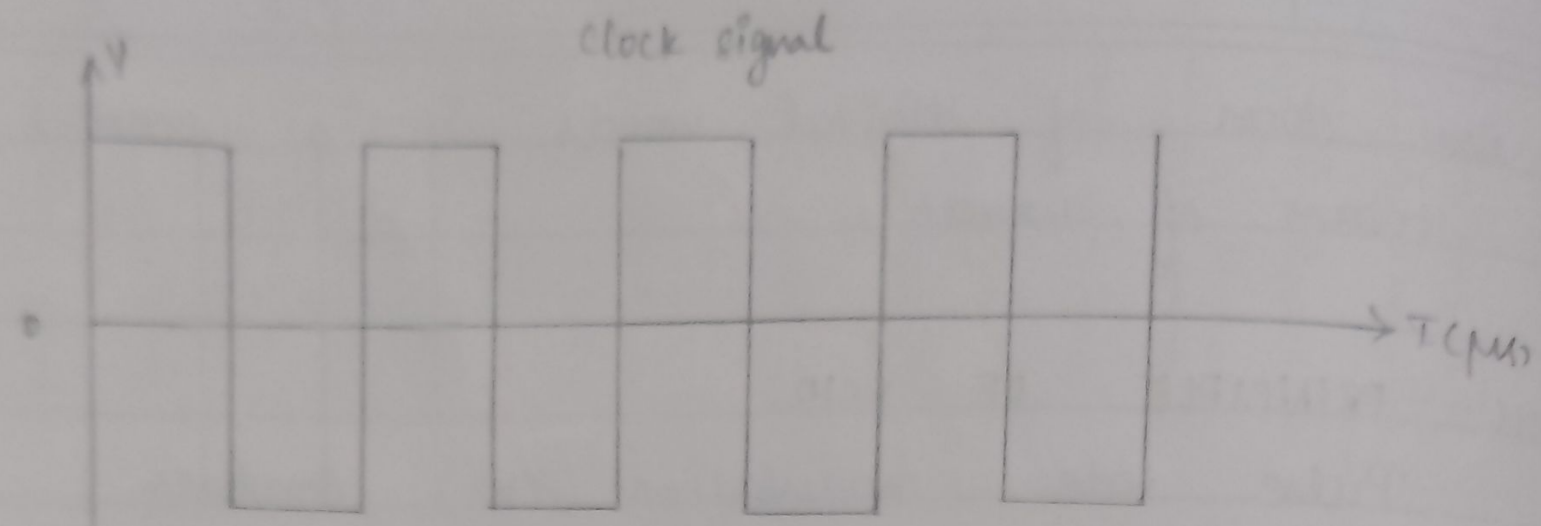
1 2 3 4

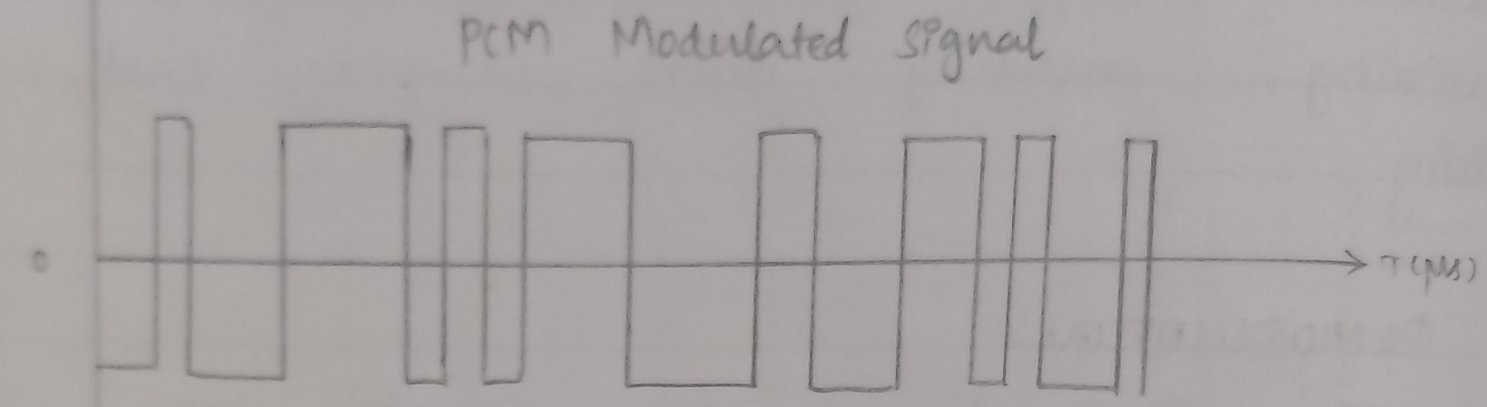
GND

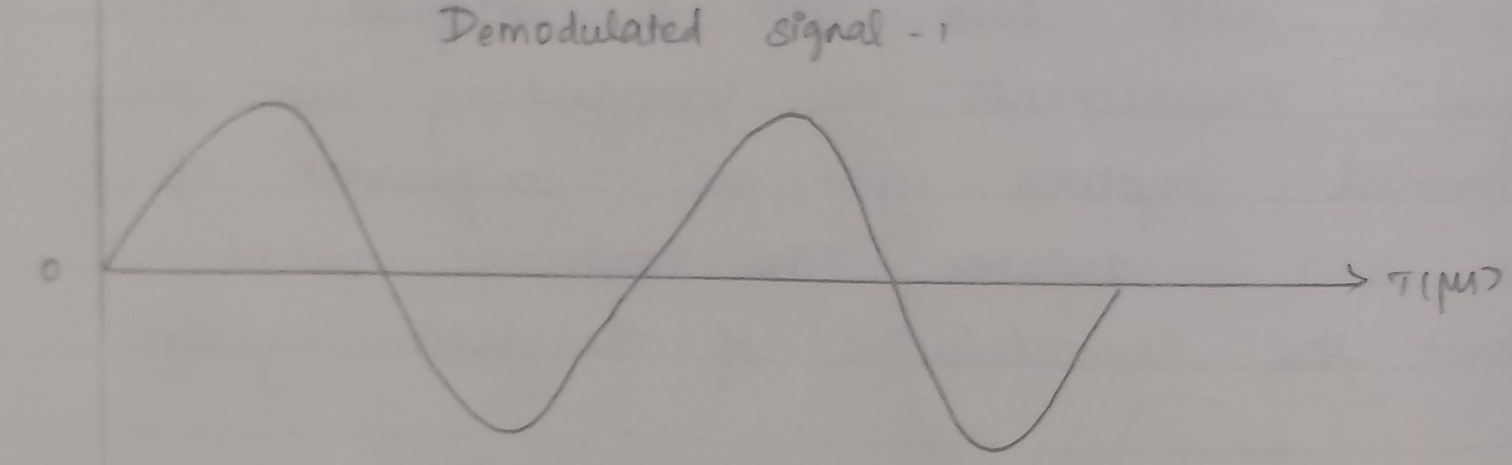
GND

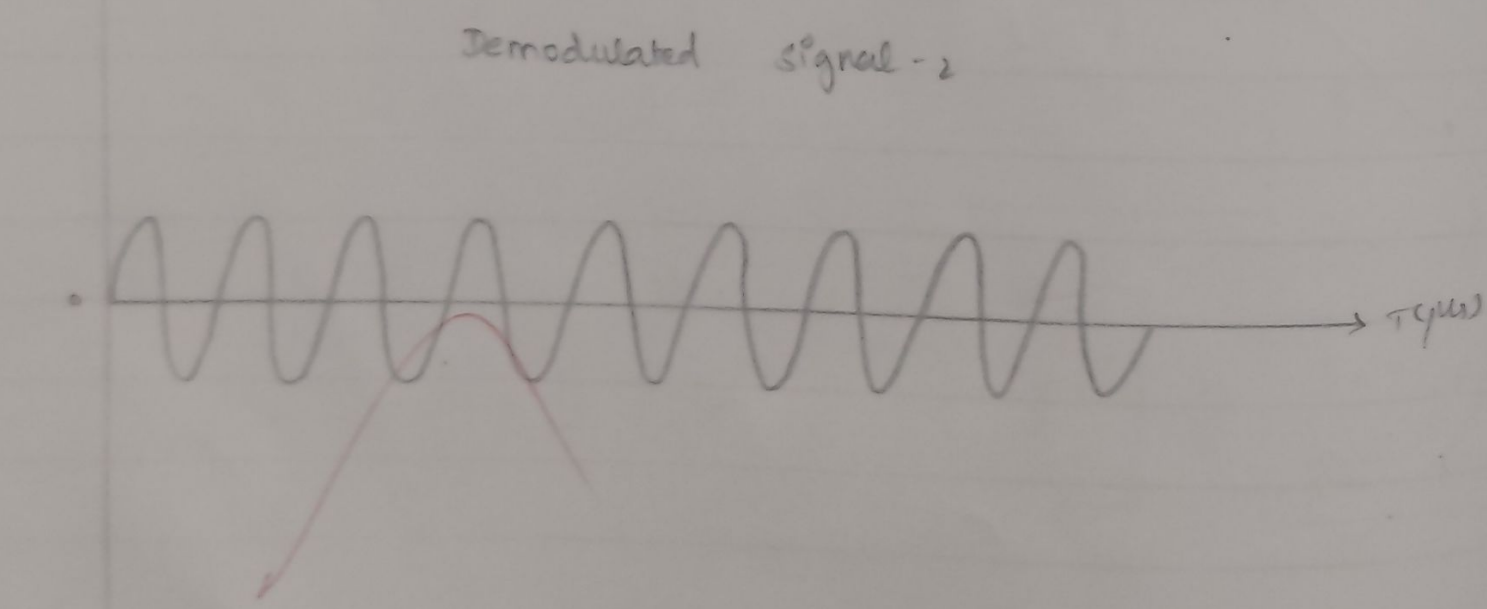
11/10/202

MODEL GRAPH:


Clock signal
V
0
T(μs)


PCM Modulated Signal
0
T(μs)

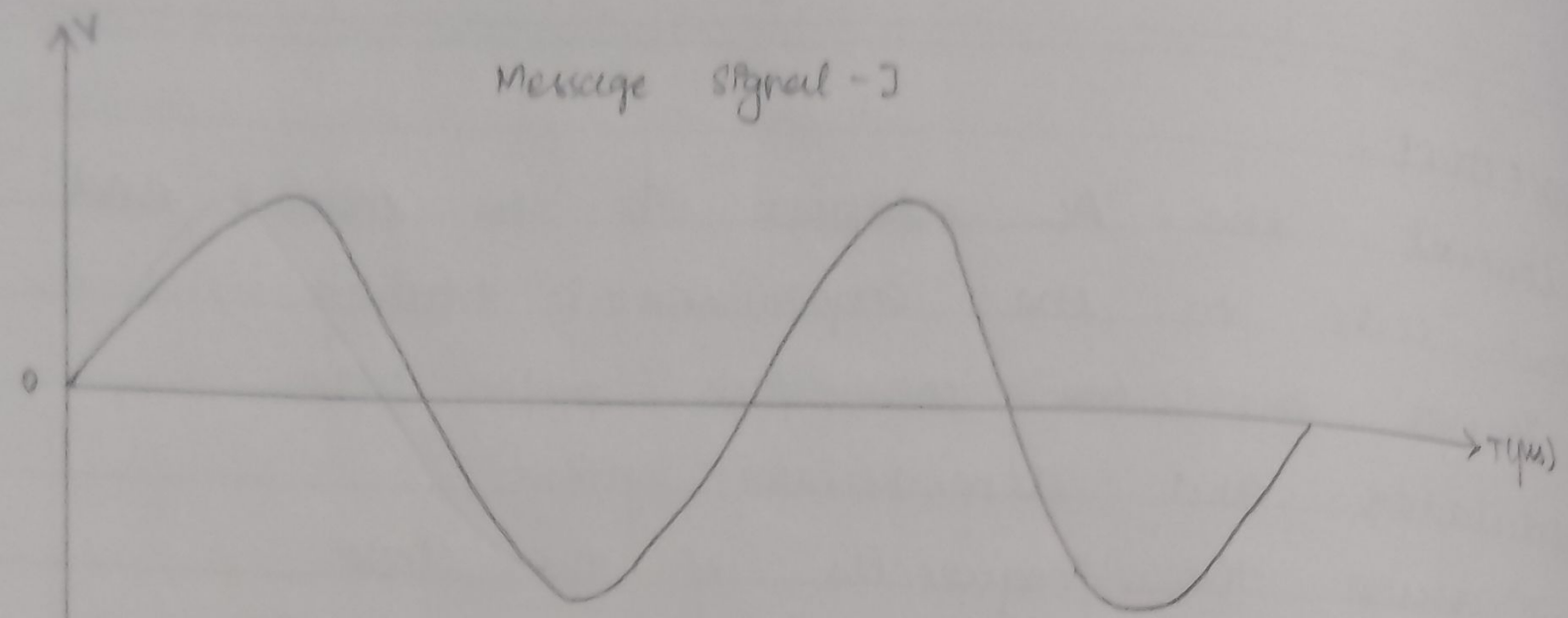

Demodulated signal - 1
0
T(μs)


Demodulated signal - 2
0
T(μs)

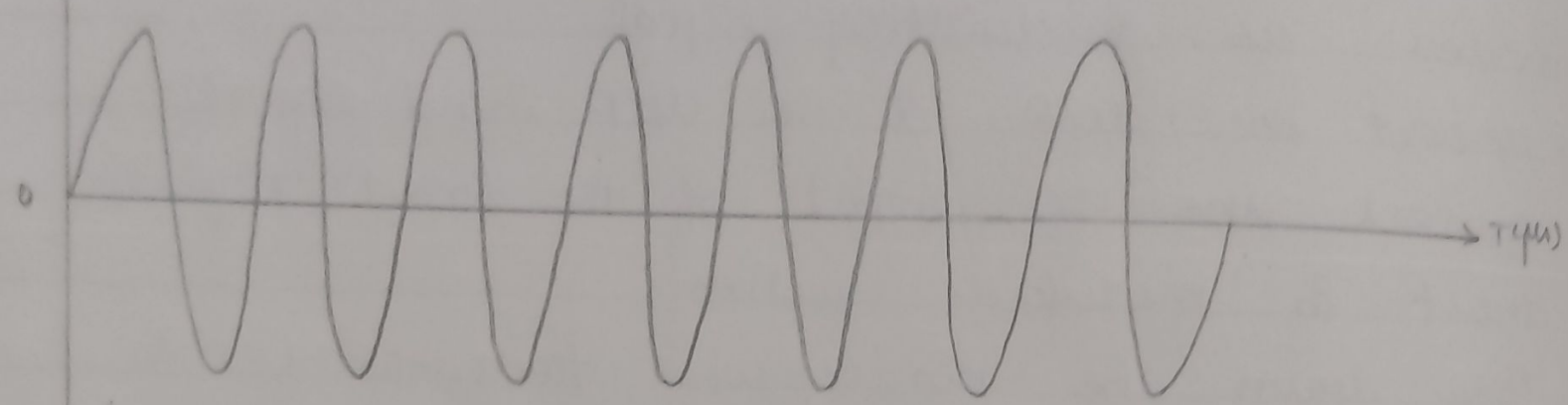
PROCEDURE :

1. Connect the AC adapter in the mains and other code to the experimental trainer
2. Power on the force - tech pulse code modulator and demodulator system
3. Measure the frequencies of the clock
4. For visual convenience, a DC variable is provided as modulating signal
5. Connect the clock in the clock input signal
6. Connect the DC input of the input signal
7. Select in modulator section
8. Note down the binary word from the LED's
9. To see the data which is transmitted through oscilloscope and serial data to channel 2.
10. Now with modulation signal to the input, observe the waveform at the output of which is in quantized level and observe the demodulated signal output

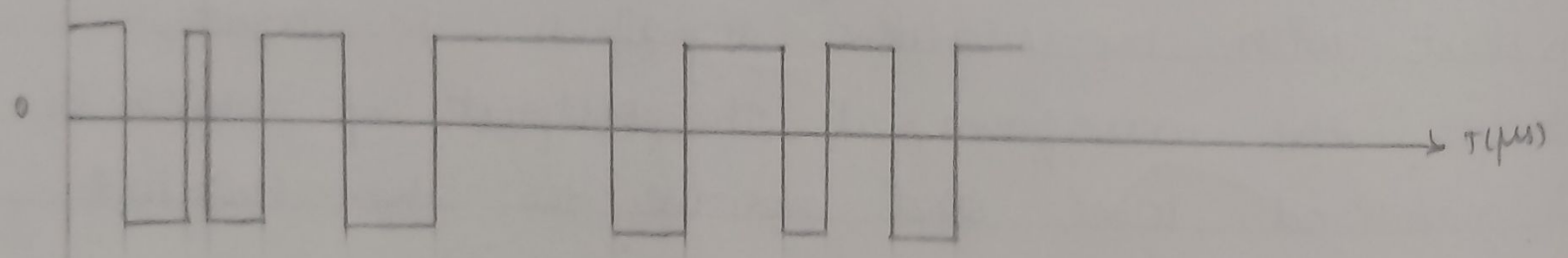
Message Signal - I



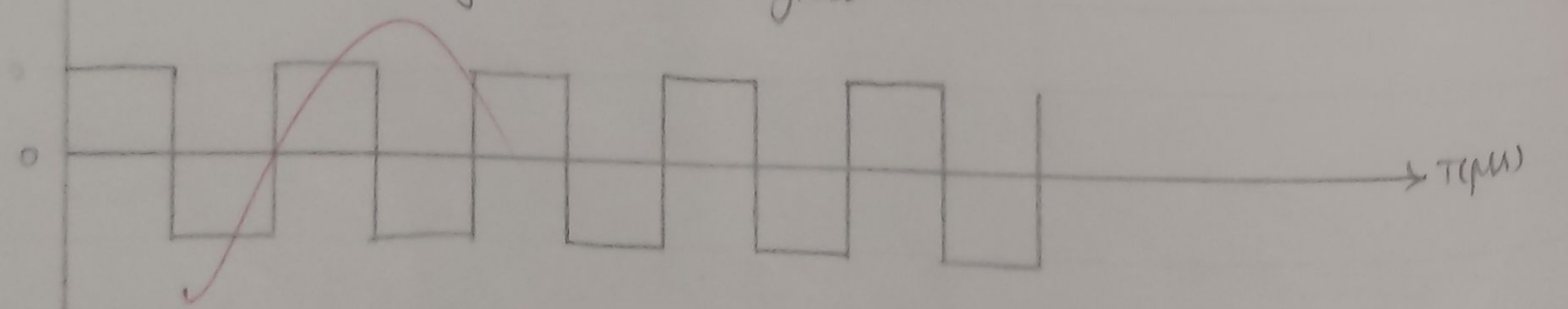
Message signal - II



PRBS Signal



Synchronous Signal

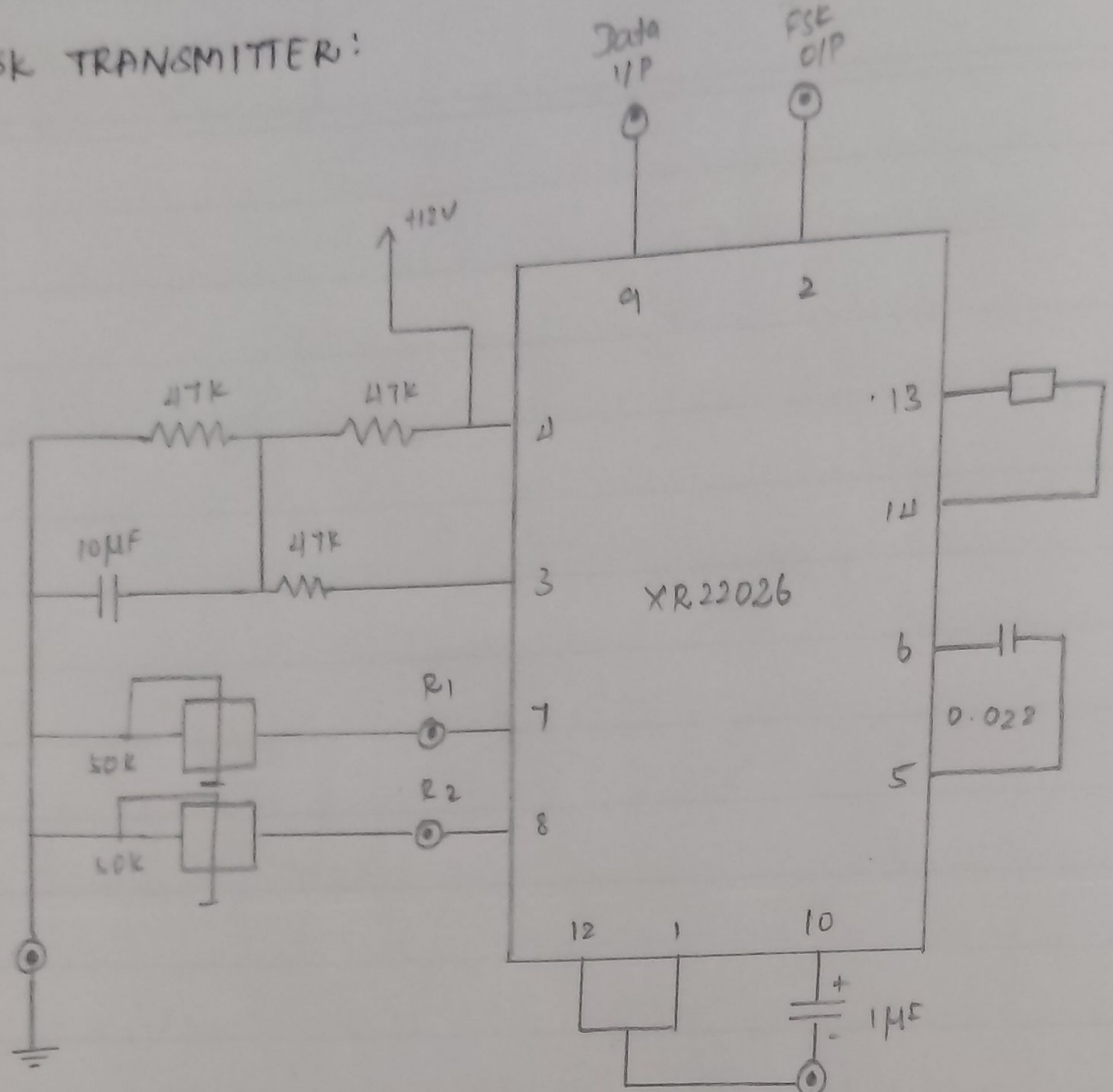


Particulars	(Max) Marks	Obtained
Observation	3	3
Output / Results	5	5
Performance /Viva-Voce	2	2
Total	10	10
Signature with date		

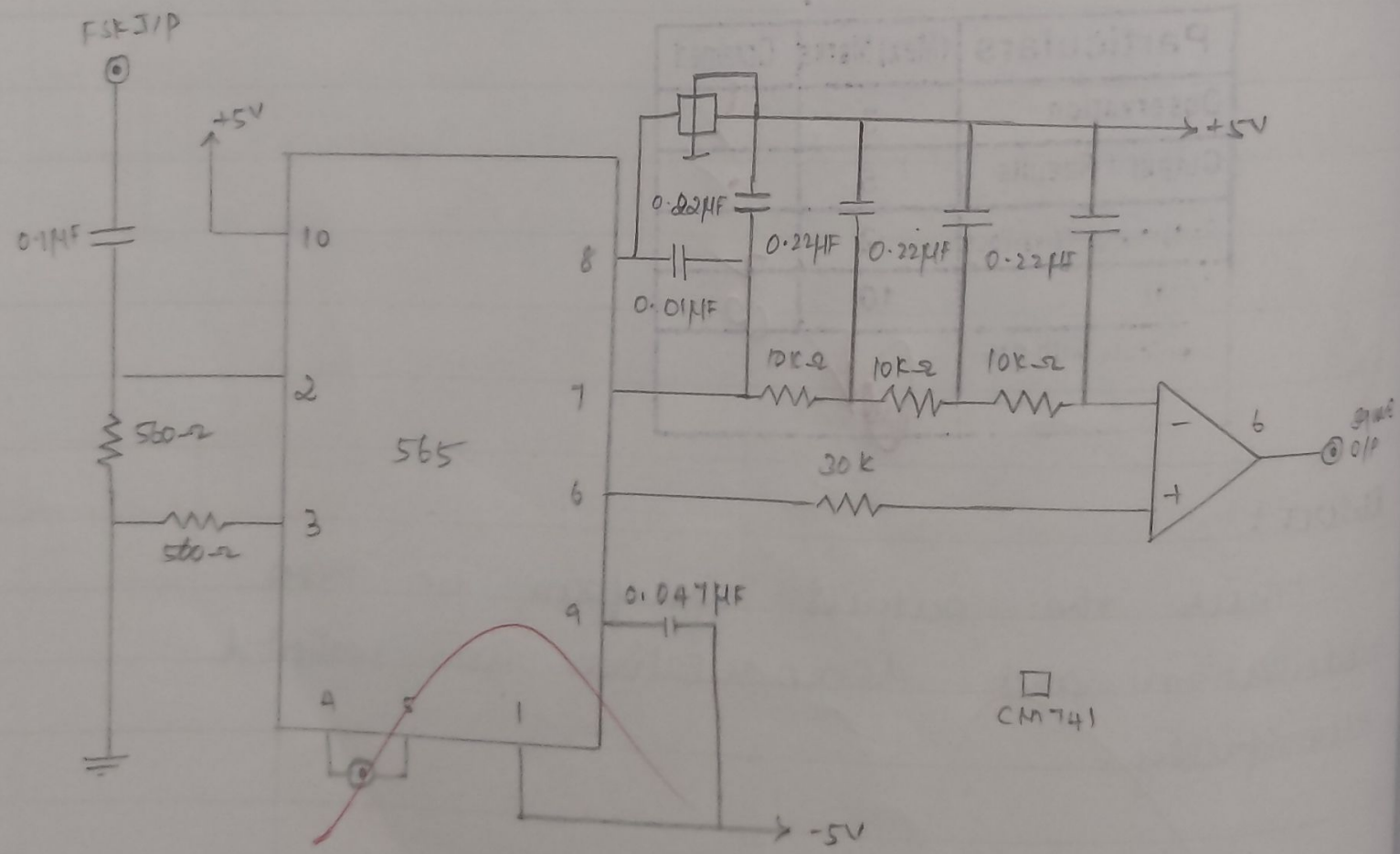
RESULT:

Thus the output waveform for PCM modulation and demodulation was verified sucessfully.

FSK TRANSMITTER:



FSK RECEIVER:



Expt. No. 13
Date 01/10/26

Page No. 113

IMPLEMENT DIGITAL MODULATION
ASK, PSK, FSK

AIM:

To study and implement digital modulation technique (ASK, PSK, FSK).

APPARATUS REQUIRED:

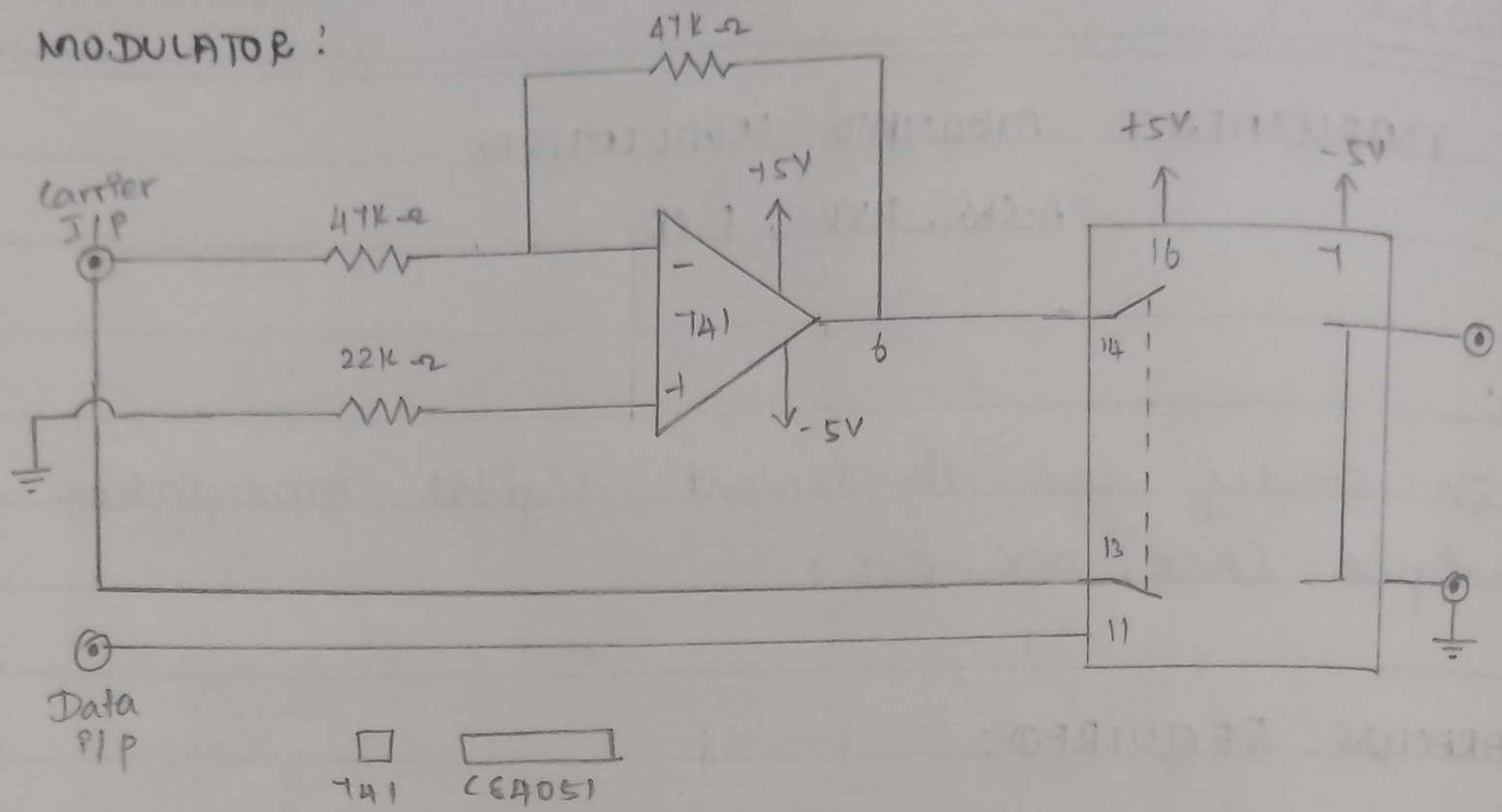
S.NO	COMPONENTS	RANGE	QUANTITY
1.	Digital modulation trainer kit.	-	1
2.	Power supply for trainer kit	-	1
3.	Power cord	-	1
4.	Patch cord	-	1
5.	DSO	(0-30) MHz	1

A) FSK MODULATION:

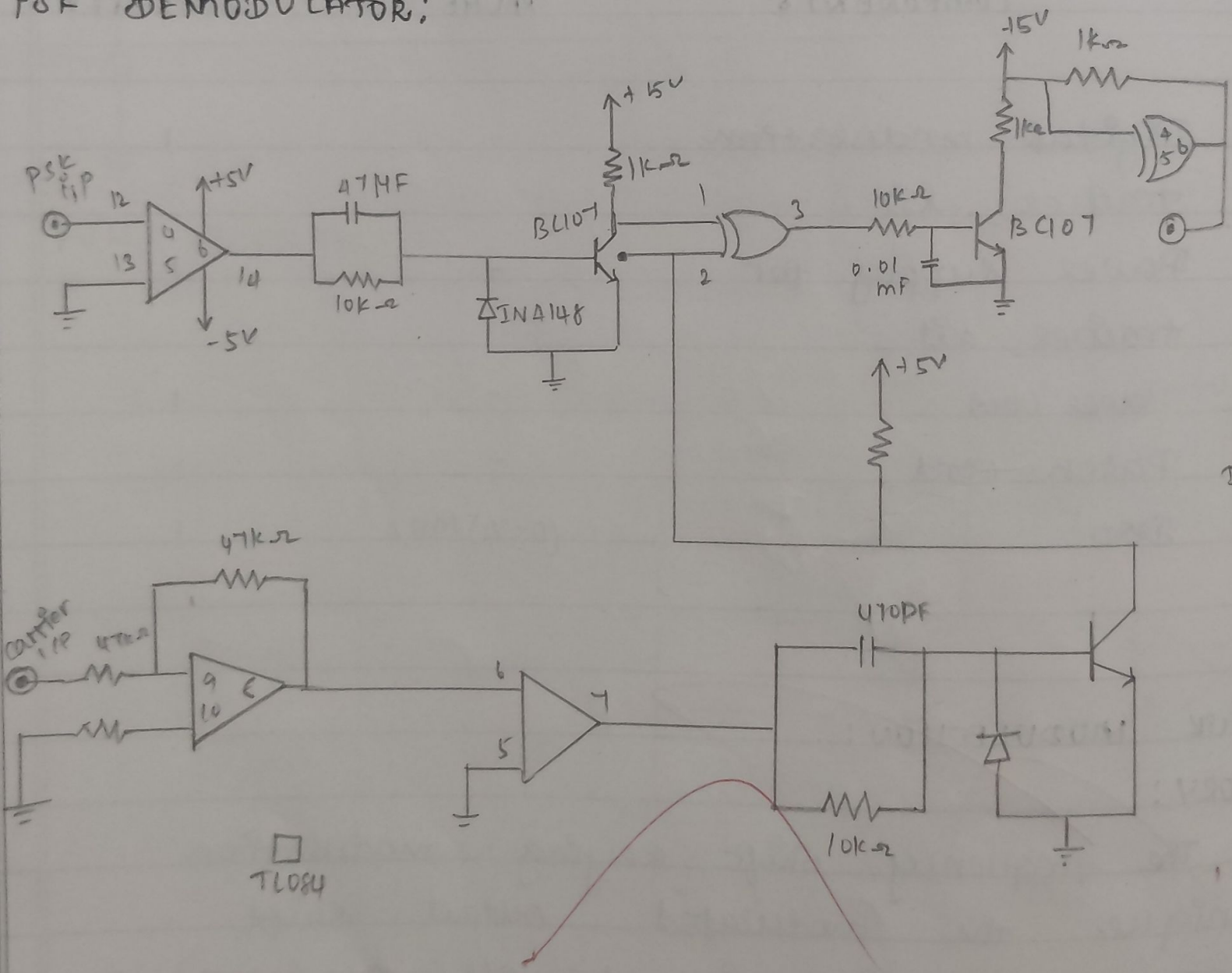
THEORY:

The frequency shift keying modulation technique the modulated output shift between two frequencies for all one (mark) to 0 (space) transitions. The carrier frequencies chosen for FSK modulation are 60KHz and 1MHz note

PSK MODULATOR :



PSK DEMODULATOR;



them mark and space frequencies and modulation is by a square wave.

PROCEDURE:

1. Connect power supply's proper polarity to the kit and switch it on.
2. Connect clock and data generated to coding clock in and data input respectively by means of connecting cords provided
3. Connect the data input to the control input of the carrier modulator kit
4. Connect carrier component sine 1 to input 1 and sine 2 to input 2 of the carrier modulator kit.

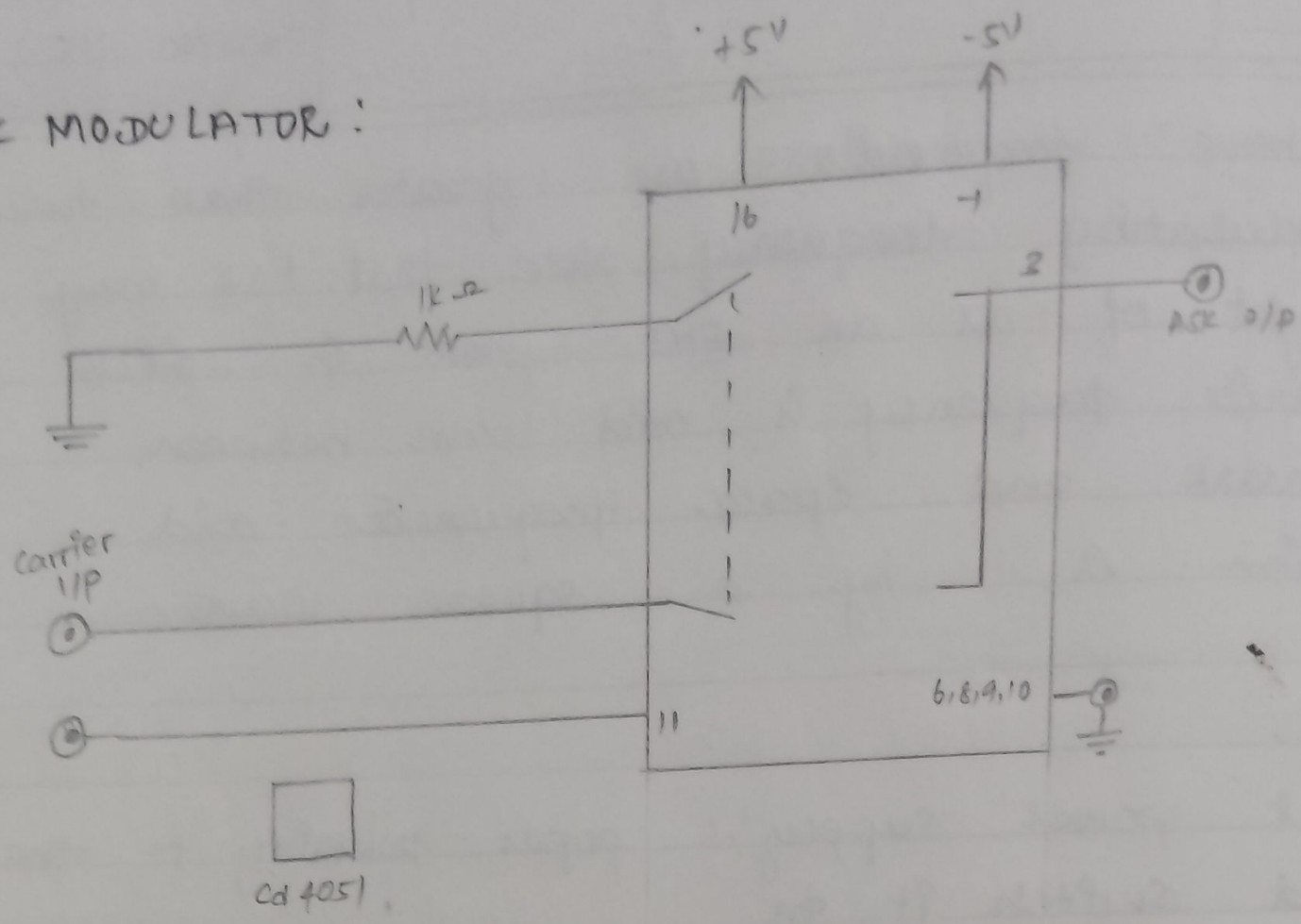
Demod o/p

B) PSK MODULATION:

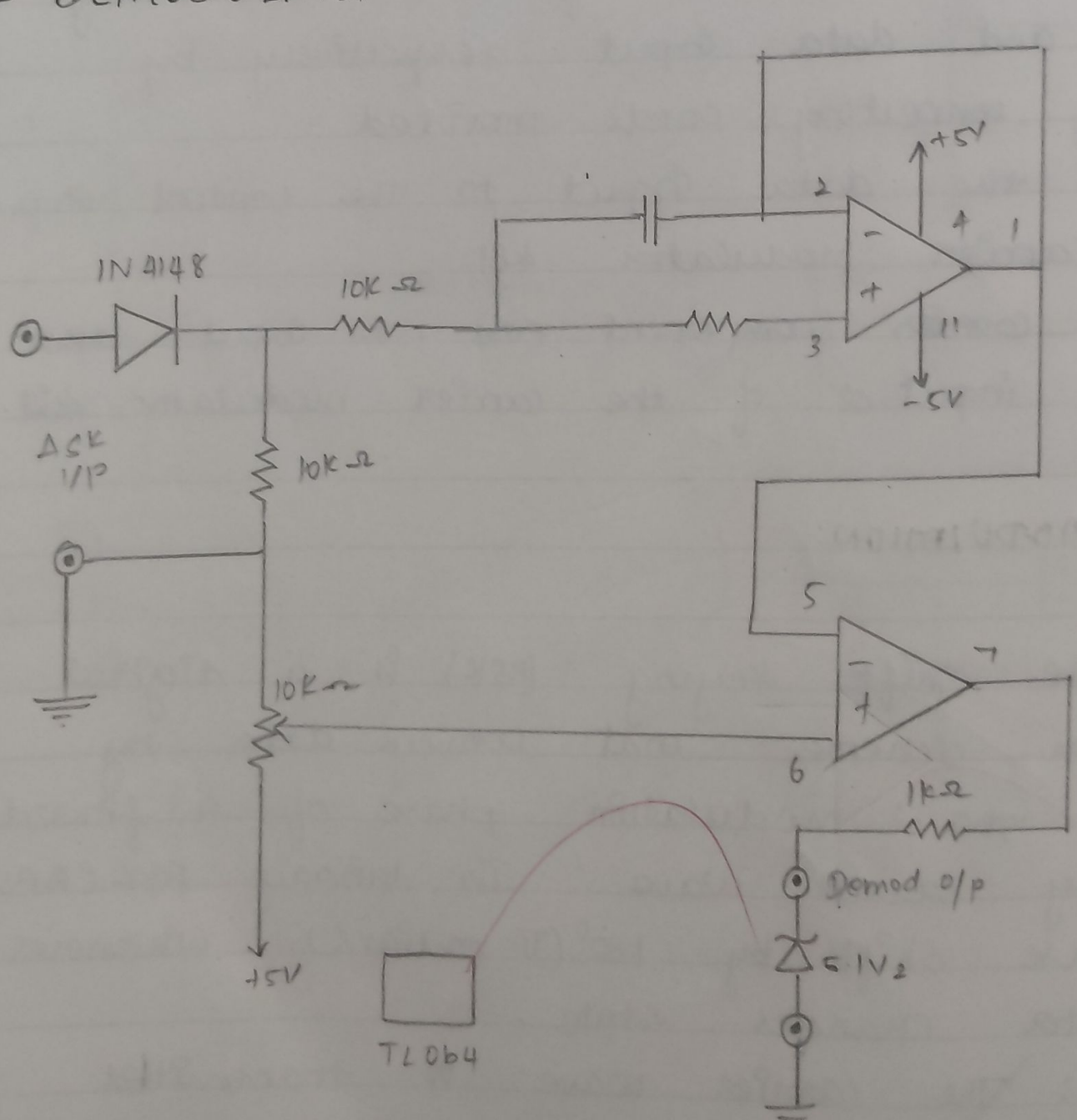
THEORY:

Phase shift keying (PSK) is a digital modulation scheme that conveys data by modulation phase of a fixed

ASK MODULATOR :



ASK DEMODULATOR :



Binary 0: The carrier wave is transmitted with 180° phase shift.

PROCEDURE:

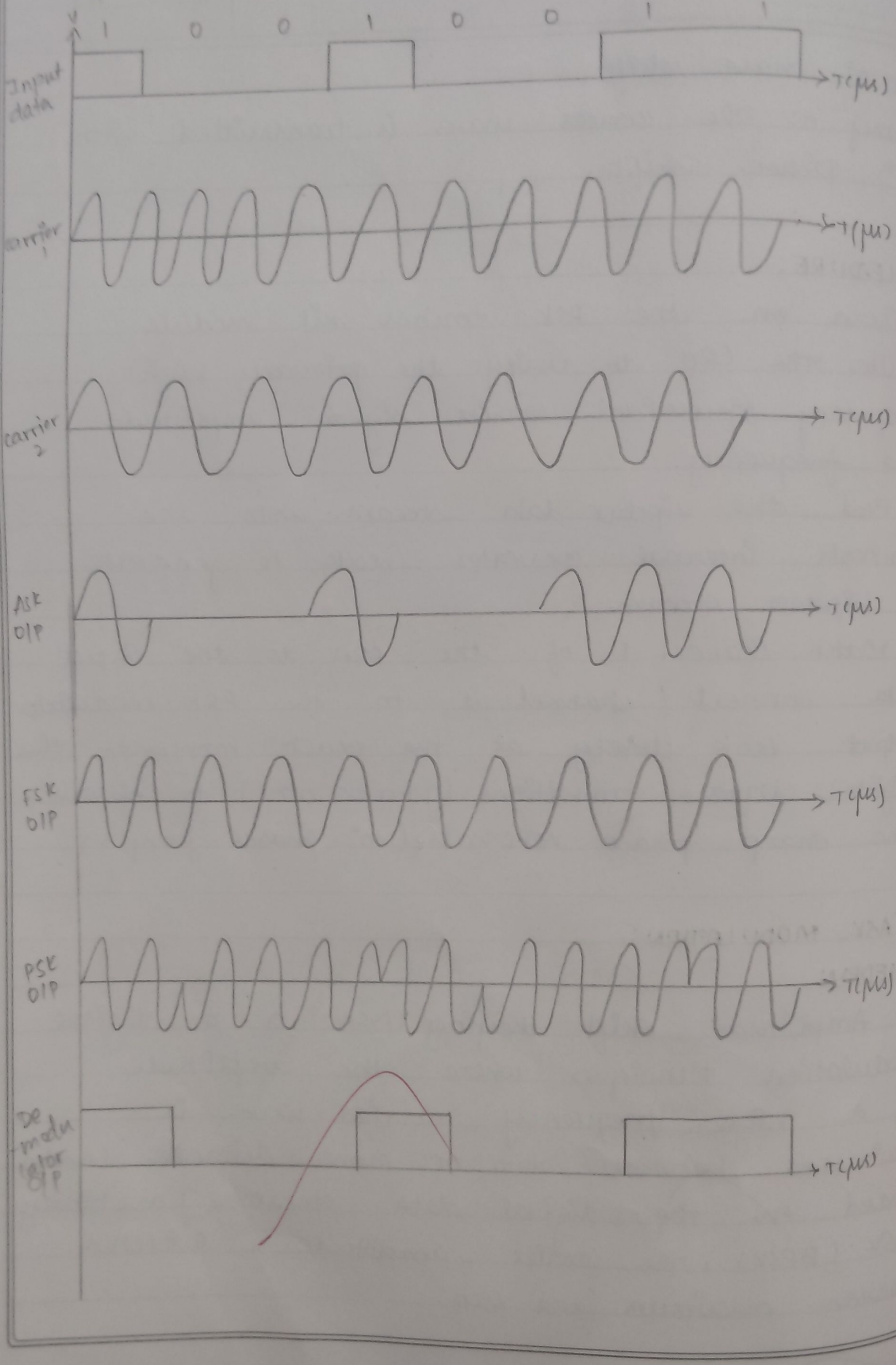
1. Turn on the PSK trainer kit module
2. Use the CRO to verify the primary high-frequency sinusoidal carrier signal amplitude and frequency.
3. Feed the bipolar data stream into the trainer's internal converter block to generate a bipolar stream
4. Match channel 1 of the CRO to the input data. Connect channel 2 to the PSK modulator output. Look closely at the exact moments the digital data transitions (1→0 or 0→1) to observe the sharp phase reversals (180° phase jumps).

C) ASK MODULATION:

THEORY:

Amplitude shift keying (ASK) is a digital modulation technique where the amplitude

MODEL GRAPH:


V
1
0
0
1
0
0
1
1
Input data
T(μs)
carrier 1
T(μs)
carrier 2
T(μs)
ASK O/P
T(μs)
FSK O/P
T(μs)
PSK O/P
T(μs)
De modulator O/P
T(μs)

Binary '1' (mark): The carrier wave is transmitted with maximum amplitude.
Binary '0' (space): The carrier wave is completely suppressed (amplitude is zero).

PROCEDURE:

1. Connect the power adapter to the trainer kit and switch it on
2. Connect channel 1 of the CRO to the carrier generated output.
3. Set the amplitude, and note down the frequency of the carrier sine wave.
4. Connect channel 1 of the CRO to the Data generator output to observe the sequential binary data sequence
5. Connect channel 2 of CRO to the ASK modulated output. Synchronise with channel 1 (Data Input) to observe the amplitude variations matching the binary bits.

Particulars	(Max) Marks	Obtained
Observation	3	3
Output / Results	5	5
Performance /Viva-Voce	2	2
Total	10	10
Sign. with date		

RESULT:

Thus the digital modulation techniques (ASK, PSK, FSK) was implemented and the output waveforms were verified successfully.